



Evaluation of Six Subgrid-Scale Models for LES of Wind Farms in Stable and Conventionally-Neutral Atmospheric Stratification

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Abstract

The performance of six subgrid-scale (SGS) models is analyzed for large-eddy simulations (LES) of wind-farm flows under stable (SBL) and conventionally-neutral (CNBL) atmospheric conditions. A precursor–concurrent technique is employed to provide fully developed turbulent inflow for simulations of a 40-turbine wind farm. Turbines are represented using the actuator-disc method, employing a baseline grid of 12 cells across the turbine diameter. The SBL precursor flow poses a challenge for LES, as it may not be able to resolve the small turbulent scales featured in this flow if the grid is coarse. For these precursor flows, the baseline grid results of all six SGS models are assessed relative to coarser and finer grids, with 6 and 45 cells across the diameter, respectively. The wall-adapting local eddy-viscosity (WALE) and Lagrangian-averaged scale-dependent dynamic (LASDD) models exhibit high grid sensitivity, while the standard Smagorinsky (Smag.), anisotropic minimum-dissipation (AMD), one-equation turbulent kinetic energy (TKE), and stability-dependent Smagorinsky (SDS) models show low sensitivity. For the wind-farm simulations conducted with the baseline grid, the AMD and SDS models predict similar wind-farm performance. In contrast, the WALE and LASDD models predict nearly 30% less power output, primarily due to their prediction of lower inflow wind speeds. CNBL simulations on the baseline grid show reduced sensitivity to the SGS model due to larger atmospheric turbulence and length scales compared to the SBL flow. Among the six models, the AMD model demonstrates ease of implementation, the least sensitivity to grid size for the SBL precursor flow, and predictions that are consistent with other models and higher-order pseudo-spectral LES solvers, making it a suitable choice for LES of wind-farm flows under both stable and conventionally-neutral conditions.

Keywords Atmospheric stability · Large-eddy simulation · Smagorinsky model · Subgrid-scale model · Wind-farm flow

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1 Introduction

The application of large-eddy simulations (LES) for analyzing wind farms and their interactions with atmospheric boundary-layer (ABL) flows has experienced substantial growth over the past decade (e.g., Calaf et al. 2010; Abkar and Porté-Agel 2013; Stevens et al. 2014; Allaerts 2016; Stieren and Stevens 2022). This increase is attributed to the ability of LES to offer detailed insights into flow physics that surpass the limitations of lower-order models and the constraints of observational data. LES solvers are capable of resolving stresses and fluxes at scales larger than the filter width, typically corresponding to the grid size, but the unresolved stresses and fluxes at scales smaller than the filter width have to be accounted for with a subgrid-scale (SGS) model. Various SGS models have been developed for atmospheric boundary-layer flows (Smagorinsky 1963; Deardorff 1980; Lilly 1992; Zang et al. 1993; Sullivan et al. 1994; Meneveau et al. 1996; Kosović 1997; Nicoud and Ducros 1999; Stevens et al. 2000; Esau 2004; Stoll and Porté-Agel 2006; Abkar and Moin 2017). These models rely on the eddy-viscosity assumption, wherein the SGS stresses are proportional to the rate of strain tensor, with the proportionality constant being the SGS eddy viscosity (Allaerts 2016). The performance of these SGS models has been evaluated and compared for atmospheric boundary-layer flows (e.g., Beare et al. 2006; Gadde et al. 2021a) and single-turbine flows (Hodgson et al. 2023; Ghobrial et al. 2024) under various atmospheric stability conditions. However, examining SGS models under flow conditions featuring wind farms has received limited attention, with studies focusing solely on wind farms interacting with pressure-driven boundary-layer (PDBL) flows (Calaf et al. 2010; Sharafabadi and Fathali 2023). In the PDBL, the flow is unidirectional, driven by a mean pressure gradient in the streamwise direction; accordingly, Coriolis effects are not considered. Moreover, this type of flow does not provide insight into the influence of thermal stratification in wind-farm flows or how these effects can best be modeled with LES (Ouro et al. 2025).

For convenience, we classify studies considering the effect of the subgrid-scale model on LES depending on three types of flow: atmospheric boundary-layer, single-turbine, and wind-farm flows. Table 1 summarizes the number of evaluated SGS models, inflow conditions, and thermal stratification for each of these studies. The inflow conditions encompass laminar uniform (LU) and turbulent uniform (TU) inflows, in addition to turbulent boundary-layer (TBL) inflow. The thermal stratification conditions comprise neutral, stable, and convective states, along with conventionally-neutral conditions, where an inversion layer separates the neutral boundary layer from the stably-stratified free atmosphere (Allaerts 2016). Table 1, to the best of our knowledge, lists all previous studies concerned with the impact of SGS models on single wind-turbine and wind-farm flows. However, from the various studies that considered impact of SGS models on atmospheric boundary-layer flows (e.g., Sullivan et al. 1994; Kosović 1997; Basu and Porté-Agel 2006; Stoll and Porté-Agel 2008; Couvreur et al. 2020), we focus our discussion to those conducted by Beare et al. (2006) and Gadde et al. (2021a).

Beare et al. (2006) simulated a moderately stable boundary-layer (SBL) flow (Holtslag 2003), specifically chosen due to its characteristic small turbulent length scales. These small scales cannot be resolved by the grid unless a sufficiently fine spacing is used; therefore, these eddies must be accounted for by the SGS model. This makes the stable boundary layer an important test case for evaluating the performance of different SGS models. Additionally, Beare et al. (2006) evaluated six SGS models: the standard Smagorinsky (Smag.) model (Smagorinsky 1963), along with its non-linear (Kosović 1997) and dynamic (Germano et al. 1991; Lilly 1992) variants, the dynamic mixed-length model (Esau 2004), the turbulent

Table 1 Summary of previous studies that have compared LES results using different SGS models for atmospheric boundary-layer, wind-turbine, and wind-farm flows

Reference	Number of SGS models	Number of turbines	Inflow type	Thermal stratification	Turbine representation	Grid size m (D/Δ)
Beare et al. (2006)	6	None	TBL	Stable	—	6.25 & 2.00
Gadde et al. (2021a)	3	None	TBL	Neutral	—	$43.6 \times 43.6 \times 6.94$
				Stable		2.08
				Convective		$10.4 \times 10.4 \times 4.16$
Sarlak et al. (2014)	2	1	LU	Neutral	ALM	(40)
Sarlak et al. (2015a)	2	1	LU & TU	Neutral	ALM	(40)
Sarlak et al. (2015b)	4	2 (2×1)	TU	Neutral	ALM	(70)
Martínez Tossas et al. (2018)	2	1	LU & TU	Neutral	ALM	$3.94 \times 3.94 \times 1.97$ (64)
Abkar (2018)	3	1	LU	Neutral	ALM	$1.63 \times 1.63 \times 0.813$ (60)
Ghobrial et al. (2024)	6	1	TBL	Stable	ALM	2.00 (64)
Hodgson et al. (2023)	2	1	TBL	Stable	ADM	DTU [(4) & (8) & (16) & (32)]
				Conventionally-neutral		EPFL [(7) & (13)]
				Convective		
Calaf et al. (2010)	2	24 (4×6)	TBL	Neutral	ADM	$24.5 \times 24.5 \times 7.81$ (12)
Sharafabadi and Fathali (2023)	4	24 (4 × 6)	TBL	Neutral	ADM	$24.5 \times 24.5 \times 7.81$ (12)
Present Study	6	None	TBL	Stable	—	15.0 & 7.50 & 2.00
				Conventionally-neutral		7.50
Present study	6	40 (10 × 4)	TBL	Stable	ADM	7.50 (12)
				Conventionally-neutral		

For the type of inflow applied, LU, TU, and TBL denote laminar uniform, turbulent uniform, and turbulent boundary-layer flows, respectively. Turbine representation methods include ALM and ADM. The last column provides the grid size ($\Delta x \times \Delta y \times \Delta z$), with z representing the vertical direction. For isotropic grids, a single value is provided. In studies evaluating SGS model sensitivity to grid size, the different sizes are separated by “&”. Additionally, in studies including turbines, the number of grid cells across the diameter is provided in parentheses

kinetic energy (TKE) model (Deardorff 1980), and a modified version of the TKE model (Sullivan et al. 1994). This selection provides a more extensive evaluation of SGS models compared to previous studies. Beare et al. also examined the sensitivity of SGS models to grid size, specifically by comparing their performance across grid sizes of 6.25 m and 2 m. For the 6.25-m grid, SGS models exhibited up to a 15% variation in mean wind speed at 100 m, the typical hub height of a turbine. The models also showed a 25% variation in predicting the height of the supergeostrophic jet (i.e., the wind maxima near the top of the boundary layer). With the finer 2-m grid, the variation in wind speed reduced to 6.5%, while the different SGS models still provided a 25% variation in the supergeostrophic jet height. This underscores the importance of selecting an appropriate SGS model when simulating a turbine under stable conditions, as a 6.5% variation in wind speed can result in a 20% difference in power output (as power is proportional to the cube of wind speed). Moreover, the height of the supergeostrophic jet dictates its interaction with wind turbines or farms, hence affecting their performance (Gadde and Stevens 2021a, b; Gadde et al. 2021b).

Another comparison of SGS models was conducted by Gadde et al. (2021a), who simulated the same stable boundary-layer case as Beare et al. (2006), using a 2-m grid. However, Gadde et al. (2021a) compared three widely used SGS models: the standard Smagorinsky, and also the anisotropic minimum-dissipation (AMD) (Verstappen 2011; Rozema et al. 2015; Abkar and Moin 2017), and Lagrangian-averaged scale-dependent dynamic (LASDD) (Stoll and Porté-Agel 2006) models. In addition to the stable boundary-layer flow, the study extended its analysis to include pressure-driven and convective boundary-layer (CBL) flows, thereby assessing the performance of these SGS models under varying stability conditions. Across all stability conditions, the comparison revealed that the mean profiles produced by the AMD and LASDD models were in closer agreement with higher-resolution data (Beare et al. 2006; Sullivan et al. 2016) compared to those produced by the Smagorinsky model.

For studies including wind turbines, Table 1 also details the number of turbines and the method used to represent their rotor, typically LES studies have been combined with either the actuator-line method (ALM) or the actuator-disc method (ADM) (Wu and Porté-Agel 2011; Martínez Tossas et al. 2012). In ALM, turbine blades are segmented, with each segment's lift and drag forces calculated using airfoil data and local wind speed, requiring about 32 segments per blade for accurate near-wake resolution (Gadde et al. 2021b). In contrast to the ALM, the ADM represents the turbine as a disc with uniform distribution of thrust force. Although ADM does not resolve individual blades, it can predict the wake shape after three diameters downstream of the turbine, provided the grid spacing provides at least 12 cells across the turbine diameter (Calaf et al. 2010; Gadde and Stevens 2019). Thus, the chosen turbine representation method largely determines the grid size, as highlighted in the last column of Table 1. The table provides the grid size ($\Delta x \times \Delta y \times \Delta z$), with z representing the vertical direction. For isotropic grids, a single spacing value is provided. In cases where studies have evaluated the sensitivity of SGS models to grid size, the different grid sizes are indicated and separated by the symbol "&". Additionally, in studies including turbines, the number of grid cells across the diameter is provided in parentheses.

For wind-turbine flows, five studies have conducted LES to analyze wind-turbine performance, subject to laminar and turbulent uniform inflows, under neutral conditions (Sarlak et al. 2014, 2015a, b; Martínez Tossas et al. 2018; Abkar 2018). For both types of inflow, these studies found that mean blade loading, power output, and near-wake structure (i.e., up to 12 diameters downstream of the turbine) showed minimal sensitivity to the SGS model. The far-wake structure was influenced by the SGS model in the case of laminar inflows. Conversely, in the case of turbulent inflows, the far-wake structure was governed by the inflow turbulence intensity, with reduced sensitivity to the SGS model (Sarlak et al. 2015a;

Martínez Tossas et al. 2018; Abkar 2018). The limited impact of the SGS model was attributed to the fine grid, which provided 64 grid cells across the turbine diameter, as required for the actuator-line method used in these studies (Sarlak et al. 2015b). On the whole, these five studies have provided useful understanding of sensitivity of wind-turbine flows to the SGS model. However, they have focused on idealized scenarios with uniform inflow and neutral conditions, overlooking thermal stratification effects.

To address this gap, Ghobrial et al. (2024) conducted LES of a single turbine in a stable boundary layer, comparing six SGS models with the turbine modeled using the actuator-line method. All models predicted similar mean blade loading and power, except LASDD, which predicted 11.1% less power. Spanwise distributions of mean velocity and turbulence intensity were also consistent up to 12 diameters downstream, except for LASDD, which showed higher recovery rates and higher turbulence intensity after 9 diameters downstream. For each SGS model, Ghobrial et al. derived the inflow conditions for the turbine simulation from a precursor boundary-layer flow simulation conducted without the turbine. The LASDD precursor simulation deviated from the others, explaining the differences in turbine performance and wake structure predicted by LASDD. Previous studies indicate that the grid size required by the actuator-line method to represent turbine blades effectively can resolve most turbulent eddies, leaving only a minor portion to be accounted for by the SGS model. This minimizes the impact of the SGS model on simulation results. However, such simulations have relatively high computational costs. In contrast, the actuator-disc method enables the use of coarser grids, with a larger portion of turbulence being modeled by the SGS model. This makes it essential to understand how the choice of SGS model and grid size impacts the LES of wind-farm flows in actuator-disc applications.

In this context, Hodgson et al. (2023) simulated a turbine using the actuator-disc method with a grid providing 13 grid cells across the diameter. The study covered three types of flow conditions: stable, conventionally-neutral (CNBL), and convective boundary layers. Two LES solvers were evaluated: DTU finite-volume solver with the turbulent kinetic energy SGS model and EPFL pseudo-spectral code with the LASDD SGS model. The stable case showed the largest deviations between the LES solvers, with a 3.8% variation in hub-height wind speed. For the conventionally-neutral case, both solvers provided similar wind speeds over the vertical extent of the turbine rotors, with a 0.7% variation at hub height. However, the finite-volume code predicted a 25% larger boundary-layer height than the pseudo-spectral code. The convective boundary-layer results from both solvers showed good agreement, with a 0.5% variation in hub-height wind speed. The mean power output varied by approximately 7.2%, 4.3%, and 3.8% between the solvers for the stable, conventionally-neutral, and convective cases, respectively. Extending the investigation of wind-turbine flow toward wind-farm flow increases the complexity due to the larger computational domains required to encompass an entire farm and its wake.

For wind-farm flows, the fine grid size required to apply the actuator-line method imposes very high computational demands and is currently impractical for the large computational domains necessary for wind-farm flows. Consequently, the actuator-disc method, which offers greater computational efficiency, is more suitable for analyzing wind-farm flows. Calaf et al. (2010) studied the interaction of a 24-turbine wind farm with a pressure-driven boundary layer using the Smagorinsky and LASDD models in two pseudo-spectral codes. The turbines were represented using the actuator-disc method. Their results indicated that the Smagorinsky model predicted higher velocity gradients near the bottom surface, resulting in increased wind speeds over the vertical extent of the turbine rotors compared to the LASDD model. For the same wind-farm setup, Sharafabadi and Fathali (2023) conducted LES using the Smagorinsky, wall-adapting local eddy-viscosity (WALE) (Nicoud and Ducros 1999), dynamic Smagorin-

sky, and Lagrangian-averaged dynamic (Meneveau et al. 1996) models, also representing the turbines as actuator discs. The WALE model exhibited higher wind speed at turbine height relative to the other models, together with predicting 8% greater power production and 10% higher wake recovery rate. They concluded that the choice of the SGS model substantially influenced estimates of velocity and turbulence profiles, as well as wake recovery, thereby impacting the farm's performance. This dependency on SGS model choice requires further investigation, particularly in stratified atmospheric stability conditions.

Drawing from the reviewed literature, the present study is driven by three motivations. First, studies investigating the impact of SGS models on wind-farm flows have, to-date, focused on farms operating within pressure-driven neutral boundary layers, which represent idealized conditions (Calaf et al. 2010; Sharafabadi and Fathali 2023). It is crucial to extend this research to include stable boundary layers, characterized by smaller length scales that present challenges for LES, and for which improved understanding of SGS model performance is required (Cañadillas et al. 2023). Stable boundary layers are also characterized by the presence of low-altitude supergeostrophic jets. The interaction between these jets and wind farms influences the flow dynamics and overall performance of the farms (Gadde and Stevens 2021a,b; Gadde et al. 2021b). Consequently, it is essential to examine the ability of various models to predict these jets and their interactions with wind farms. Moreover, conventionally-neutral boundary layers should also be considered, as they are typical of offshore conditions, which hold substantial interest for the wind energy sector. Second, previous studies of wind-farm flows have tested the Smagorinsky model, including its dynamic versions, and the wall-adapting local eddy-viscosity model. However, there is a need to explore other models, such as the one-equation turbulent kinetic energy model and the anisotropic minimum-dissipation models, which are increasingly used in LES of wind-turbine and wind-farm flows. Finally, previous studies have provided valuable diagnostics through mean vertical profiles of flow variables, wake profiles, and overall budget analyses of the wind farm. However, it is beneficial to expand these diagnostics, specifically through examination of the streamwise development of the flow along and downstream of the farm and energy budget analysis for each row of turbines. These can offer a more comprehensive understanding of the SGS models impact on the interaction between the wind farm and the atmospheric flow, as well as on the performance of individual turbine rows.

Therefore, the present study aims to quantify the sensitivity of LES of atmospheric boundary-layer and wind-farm flows to SGS model choice under stable and conventionally-neutral conditions. Six SGS models are evaluated: the standard Smagorinsky, wall-adapting local eddy-viscosity, anisotropic minimum-dissipation, one-equation turbulent kinetic energy, stability-dependent Smagorinsky and Lagrangian-averaged scale-dependent dynamic models. The sensitivity of LES results to these SGS models must be established for grid spacings relevant to wind-farm flow simulations, particularly those meeting the resolution requirements of the widely adopted ADM. The SBL flow, which provides inflow conditions for the wind farm in the stable case, poses a particular challenge for LES due to its low atmospheric turbulence. Accordingly, this study also evaluates SBL flow results from different SGS models on a baseline grid (satisfying ADM resolution requirements) and compares these to finer and coarser grids. The impact of the SGS model on LES results is analyzed by assessing the interaction between the wind farm and atmospheric boundary-layer flow, as well as the kinetic energy budget, to elucidate the flow mechanisms driving wind-farm performance.

The structure of the study is as follows: Section 2 presents the details of the in-house LES solver including a summary of the evaluated SGS models; Sect. 3 describes the key flow parameters used to generate the boundary-layer flows, the wind-farm layout, the com-

putational domain, and the diagnostics monitored to compare the models; Sect. 4 presents the results of the atmospheric boundary-layer flows and wind-farm simulations for SBL and CNBL cases; Sect. 5 provides a critical discussion of the results together, along with a number of recommendations; and Sect. 6 concludes this article.

2 Large-Eddy Simulations

The in-house digital offshore FArm simulator (DOFAS) (Ouro et al. 2019) is used to conduct LES of atmospheric boundary-layer flows with wind farms. DOFAS, which is based on FORTRAN-90 and fully parallelized with MPI (Cevheri et al. 2016; Ouro et al. 2021), has been extensively used in studying tidal stream turbines (Ouro et al. 2017, 2024; Willden et al. 2023) and arrays (Ouro et al. 2019, 2023; Ouro and Nishino 2021), as well as wind turbines operating in stable conditions (Ghobrial et al. 2024). DOFAS uses a staggered grid to store the flow variables with second-order central-difference discretization in all spatial directions (Griebel et al. 1998a). Time integration is performed using a modified three-step Runge–Kutta scheme (Sullivan et al. 1996) and an algebraic multi-grid method is used to solve the Poisson pressure equation (Griebel et al. 1998b). DOFAS solves the spatially-filtered set of the incompressible Navier–Stokes equations (e.g., Abkar and Porté-Agel 2013; Allaerts 2016; Gadde and Stevens 2021a) under the Boussinesq approximation for mass, momentum, and energy conservation:

$$\frac{\partial \tilde{u}_i}{\partial x_i} = 0; \quad (1)$$

$$\frac{\partial \tilde{u}_i}{\partial t} + \frac{\partial (\tilde{u}_j \tilde{u}_i)}{\partial x_j} = f_c \varepsilon_{ij3} (\tilde{u}_j - U_{Gj}) - \frac{1}{\rho_o} \frac{\partial \tilde{P}^*}{\partial x_i} + \delta_{i3} g \beta (\tilde{\theta} - \langle \tilde{\theta} \rangle) - \frac{\partial \tau_{ij}^{SGS}}{\partial x_j} + f_i; \quad (2)$$

$$\frac{\partial \tilde{\theta}}{\partial t} + \frac{\partial (\tilde{u}_j \tilde{\theta})}{\partial x_j} = - \frac{\partial q_j^{SGS}}{\partial x_j}, \quad (3)$$

where the operation $\tilde{\cdot}$ represents LES filtering; $\langle \cdot \rangle$ denotes horizontal averaging operation; u_i and x_i are the velocity components and spatial coordinates, respectively, with $i = 1, 2$ being the horizontal directions and $i = 3$ being the vertical direction (throughout this study (x, y, z) and (u, v, w) are used to represent (x_1, x_2, x_3) and (u_1, u_2, u_3) , respectively); t is time; f_c is Coriolis parameter and ε_{ijk} is the Levi–Civita tensor; U_{Gi} are the velocity components of the geostrophic wind. Notably, the geostrophic wind speed is related to the mean pressure gradient driving the flow through the geostrophic balance law, applicable above the boundary layer, as follows (Allaerts 2016; Gadde and Stevens 2021a):

$$U_{G1} = \frac{-1}{\rho_o f_c} \frac{\partial P_\infty}{\partial x_2}; \quad U_{G2} = \frac{1}{\rho_o f_c} \frac{\partial P_\infty}{\partial x_1}, \quad (4)$$

where ρ_o is the reference density and P_∞ is the mean background pressure. The filtered modified pressure is denoted by $\tilde{P}^*$ and computed as:

$$\tilde{P}^* = \tilde{P} + \rho_o \tau_{kk}^R / 3, \quad (5)$$

where $\tilde{P}$ is the filtered pressure perturbation and $\tau_{kk}^R / 3$ is the trace of the residual-stress tensor, which is denoted by τ_{ij}^R and given as:

$$\tau_{ij}^R = \widetilde{u_i u_j} - \tilde{u}_i \tilde{u}_j. \quad (6)$$

Kronecker's delta function is denoted by δ_{ij} ; θ is the potential temperature; g is the gravitational acceleration; $\beta = 1/\theta_o$ is the buoyancy parameter, where θ_o is the reference temperature. The SGS stress tensor (i.e., anisotropic part of the residual-stress tensor) is denoted by $\tau_{ij}^{SGS} = \tau_{ij}^R - \delta_{ij}\tau_{kk}^R/3$, and $q_j^{SGS} = u_j\theta - \tilde{u}_j\tilde{\theta}$ is the residual-heat flux. Both τ_{ij}^{SGS} and q_j^{SGS} are modeled using an SGS model (details provided in Sect. 2.1 and Appendix B).

The bottom-wall boundary condition is set according to the Monin–Obukhov similarity theory (Moeng 1984), which provides the wall shear stress as $\tau_w = \rho_o u_*^2$, with components in the horizontal plane:

$$\tau_{13}^w = \tau_w \left(\frac{\widehat{u}_1(z_1)}{M_1} \right); \quad \tau_{23}^w = \tau_w \left(\frac{\widehat{u}_2(z_1)}{M_1} \right), \quad (7)$$

where $z_1 = \Delta z/2$ is the height of the center the first grid cell from the wall, where Δx , Δy , and Δz denote the grid cell size in the x , y , and z directions, respectively; and $M_1 = \left(\widehat{u}^2(z_1) + \widehat{v}^2(z_1) \right)^{0.5}$ is the horizontal velocity magnitude at $z = z_1$ [$\widehat{\cdot}$ denotes local horizontal averaging with a filter width 2Δ , where $\Delta = \sqrt[3]{\Delta x \Delta y \Delta z}$ (Bou-Zeid et al. 2005)]. The heat flux q_w is calculated as $q_w = -\theta_* u_*$, where u_* and θ_* are the friction velocity and surface-layer temperature scale, respectively, which are obtained as:

$$u_* = \frac{\kappa M_1}{\ln\left(\frac{z_1}{z_o}\right) - \Psi_m(\zeta_1)}; \quad (8)$$

$$\theta_* = \frac{\kappa(\theta_1 - \theta_s)}{\ln\left(\frac{z_1}{z_o}\right) - \Psi_h(\zeta_1)}, \quad (9)$$

where $\theta_1 = \widehat{\theta}(z_1)$ is the potential temperature at the first grid cell from the wall; $\kappa = 0.41$ is the von Kármán constant; z_o denotes the surface roughness height; and θ_s represents the surface potential temperature. The correction functions Ψ_m and Ψ_h are used to account for thermal stability as they depend on the stability parameter $\zeta_1 = z_1/L_o$, with L_o denoting the Obukhov length defined as $L_o = -\theta_o u_*^3/(\kappa g q_w)$. Because ζ_1 is a function of u_* and θ_* , Eqs. 8 and 9 are implicit expressions for the friction velocity and the surface-layer temperature scale. The solution to this set of equations follows the methodology outlined by Allaerts (2016).

To simulate a finite wind farm with turbulent ABL inflow conditions, a precursor-concurrent technique is adopted (Stevens et al. 2014). A precursor (boundary-layer flow) simulation with no turbines (Fig. 1a) is run first. This precursor simulation is driven by a mean pressure gradient that is related to the geostrophic wind speed (Eq. (4)), with periodic boundary conditions applied in both horizontal directions. The simulation runs until the flow reaches a quasi-steady state [i.e., the root mean square of the time-averaged wind speed over two consecutive hours is less than 0.1 m/s (Beare et al. 2006)], denoted by t_{SS} . Over this period (i.e., $0 \leq t \leq t_{SS}$), a mean wind-angle controller is used to rotate the entire flow field such that the mean flow is aligned with the wind farm at hub height (Sescu and Meneveau 2014; Allaerts 2016). The flow variables from the precursor simulation during the interval $t_{SS} \leq t \leq t_F$, with t_F being the final simulation time, are used as inflow conditions for the concurrent wind-farm simulation. The concurrent simulation is initialized with the instantaneous flow field from the precursor simulation obtained at t_{SS} . The total concurrent simulation time is $t_F - t_{SS}$. A fringe region (Fig. 1b) is applied in the concurrent simulation to drive it using the precursor flow field as an inflow. The fringe forces f_i^{frx} and f_i^{fry} are

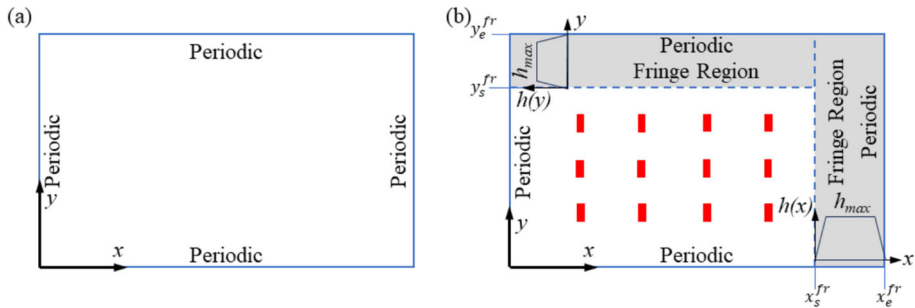


Fig. 1 Schematic showing top view of the **a** precursor domain and **b** concurrent domain, with rectangles denoting wind turbines

volume forces added in the fringe regions (Lanzilao and Meyers 2022) to impose the inflow conditions in the x and y directions, respectively.

Irregular topography and spatial variations in forcing, such as those from wind farms, can induce atmospheric gravity waves by causing vertical displacement of the flow. These gravity waves propagate upward and are advected downstream by the flow (Allaerts 2016). To prevent these waves from reflecting back into the region of interest and causing numerical errors or instabilities in computational simulations, a Rayleigh-damping layer is applied at the upper boundary of the computational domain (Klemp and Lilly 1978; Klemp and Durran 1983; Lanzilao and Meyers 2022). This layer effectively absorbs the gravity waves, ensuring the accuracy and stability of the simulation.

The forces exerted by one or more wind turbines can be resolved in DOFAS either with the actuator-line method (Yang et al. 2014; Ouro et al. 2019, 2023; Willden et al. 2023) or actuator-disc method, as in the present study. The actuator-disc method has been used in similar wind-farm simulation studies, proving accurate to predict the far-wake structure behind turbines (Calaf et al. 2010; Gadde and Stevens 2021a; Stieren and Stevens 2022; Sharafabadi and Fathali 2023). Hence, in DOFAS, the external body force f_i acting on the fluid results from the sum of these various sources, such that:

$$f_i = f_i^{wc} + f_i^{frx} + f_i^{fry} + f_i^{rd} + f_i^{turb}, \quad (10)$$

where f_i^{wc} , f_i^{frx} , f_i^{fry} , f_i^{rd} , and f_i^{turb} denote the mean wind-angle controller, fringe forces in the x and y directions, Rayleigh-damping forces, and turbine forces, respectively. Finally, Eq. (10) provides a general case when all forces are applied. However, it is important to note that f_i^{wc} is used only during the precursor simulation until a steady state is achieved, after which it is set to zero. The fringe, Rayleigh damping, and turbine forces are applied only in the wind-farm simulation. The source term of each force is applied within its designated region; for example, fringe forces are active solely within the fringe region and are set to zero elsewhere. Appendix A provides full details about the calculation of these terms.

2.1 Subgrid-Scale Models

In the present study, six subgrid-scale models are compared: the standard Smagorinsky (Smag.), wall-adapting local eddy-viscosity (WALE), anisotropic minimum-dissipation (AMD), one-equation turbulent kinetic energy (TKE), stability-dependent Smagorinsky (SDS) and Lagrangian-averaged scale-dependent dynamic (LASDD) models. These models

Table 2 Summary of the six SGS models evaluated in the present study (Smag., WALE, AMD, TKE, SDS, and LASDD), highlighting their treatment of the SGS Prandtl number and providing further remarks

SGS model	SGS Prandtl number	Remarks
Smag.	Constant	Simplest to implement. Model coefficients must be prescribed and tuned.
WALE	Constant	Simple to implement. Model coefficients must be prescribed and tuned.
AMD	Dynamic	Simple to implement. No near-wall treatment required. Provides three-dimensional variation of eddy viscosity. Turns off in laminar flow. No parameter tuning required.
TKE	Dynamic	Transport equation solved for SGS kinetic energy.
SDS	Dynamic	Avoids solving a transport equation. Assumes a balance between production, dissipation, and buoyancy terms. Turned off if Richardson number exceeds a certain value.
LASDD	Dynamic	Filtering, averaging along pathlines, and clipping required.

are eddy-viscosity models assuming there is an analogy between the effect of the SGS stresses and viscous stresses on the mean flow field. Accordingly, the SGS stresses are assumed proportional to the strain rate tensor $\tilde{S}_{ij}$, and the constant of proportionality is the SGS eddy viscosity (Allaerts 2016). Hence, the SGS stresses can be written as:

$$\tau_{ij}^{SGS} = -2\nu_{SGS}\tilde{S}_{ij}, \quad (11)$$

where ν_{SGS} is the SGS eddy viscosity, and $\tilde{S}_{ij}$ is the filtered strain rate tensor given by:

$$\tilde{S}_{ij} = \frac{1}{2} \left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} \right). \quad (12)$$

In thermally stratified flows, eddy-viscosity models assume the SGS heat fluxes to be proportional to the temperature gradients, and the constant of proportionality is the SGS eddy diffusivity. The SGS heat flux is defined as:

$$q_j^{SGS} = -\Gamma_{SGS} \frac{\partial \tilde{\theta}}{\partial x_j}, \quad (13)$$

where Γ_{SGS} is the SGS eddy diffusivity ($\Gamma_{SGS} = \nu_{SGS}/Pr_{SGS}$); and Pr_{SGS} is the SGS Prandtl number defined as the ratio between the SGS eddy viscosity and eddy diffusivity. Pr_{SGS} is prescribed as a constant in some SGS models, while others calculate a local SGS Prandtl number varying in space and time.

To prevent any numerical instability, the eddy viscosity and eddy diffusivity are constrained to non-negative values, to avoid inducing any backward cascade. Table 2 provides a summary of the six evaluated SGS models, highlighting key aspects including their treatment of the SGS Prandtl number. Further details of the six SGS models are provided in Appendix B.

In LES, the grid should be capable of resolving more than 80% of the turbulent kinetic energy, while the rest (i.e., less than 20%) should be accounted for by the SGS model (Pope 2000). In the present study, this is quantified by ρ_{TKE} , which denotes the ratio of the resolved turbulent kinetic energy E_{RES} to the total turbulent kinetic energy $E_{TOT} = E_{RES} + E_{SGS}$

over the boundary-layer height, where E_{SGS} is the SGS kinetic energy. E_{RES} is calculated as:

$$E_{RES} = 1/(2[z_i - (z_1 + \Delta z)]) \int_{z_1 + \Delta z}^{z_i} \left(\langle \overline{u'u'} \rangle + \langle \overline{v'v'} \rangle + \langle \overline{w'w'} \rangle \right) dz, \quad (14)$$

where $u'_i = \tilde{u}_i - \langle \tilde{u}_i \rangle$; $\bar{\cdot}$ denotes time averaging operation; and z_i is the boundary-layer height [i.e., the height at which the mean total shear stress first reaches 5% of its near-wall value ($\langle \bar{\tau}_w \rangle$), followed by linear extrapolation to the height at which the mean total shear stress vanishes (Kosović and Curry 2000; Beare et al. 2006; Gadde and Stevens 2021a; Bon et al. 2024)]. E_{SGS} is estimated based on the formula proposed by Deardorff (1980) (Eq. (43) in Appendix B) as follows:

$$E_{SGS} = 1/(C_k^2 \Delta^2 [z_i - (z_1 + \Delta z)]) \int_{z_1 + \Delta z}^{z_i} \langle \bar{v}_{SGS}^2 \rangle dz, \quad (15)$$

where C_k is the SGS model coefficient. In the present analysis, the first grid point from the wall is excluded from the integrations because the near wall dynamics are modeled by the wall function rather than being resolved by the grid.

To calculate the boundary-layer height z_i in Eqs. (14) and (15), the mean total shear stress $\langle \bar{\tau} \rangle = \left(\langle \bar{\tau}_{xz} \rangle^2 + \langle \bar{\tau}_{yz} \rangle^2 \right)^{0.5}$ is calculated from its resolved and SGS parts as:

$$\langle \bar{\tau}_{xz} \rangle = \langle \overline{u'w'} \rangle + \langle \bar{\tau}_{xz}^{SGS} \rangle; \quad \langle \bar{\tau}_{yz} \rangle = \langle \overline{v'w'} \rangle + \langle \bar{\tau}_{yz}^{SGS} \rangle. \quad (16)$$

Moreover, the horizontal and time-averaged resolved $\langle \bar{\tau}^{RES} \rangle$ and SGS $\langle \bar{\tau}^{SGS} \rangle$ shear stresses are given by:

$$\langle \bar{\tau}^{RES} \rangle = \left(\langle \overline{u'w'} \rangle^2 + \langle \overline{v'w'} \rangle^2 \right)^{0.5}; \quad \langle \bar{\tau}^{SGS} \rangle = \left(\langle \bar{\tau}_{xz}^{SGS} \rangle^2 + \langle \bar{\tau}_{yz}^{SGS} \rangle^2 \right)^{0.5}. \quad (17)$$

Analogously, the mean total vertical heat flux $\langle \bar{q} \rangle$ is calculated as:

$$\langle \bar{q} \rangle = \langle \overline{w'\theta'} \rangle + \langle \bar{q}_z^{SGS} \rangle, \quad (18)$$

where $\theta' = \tilde{\theta} - \langle \tilde{\theta} \rangle$; and $\langle \overline{w'\theta'} \rangle$ and $\langle \bar{q}_z^{SGS} \rangle$ correspond to the resolved $\langle \bar{q}^{RES} \rangle$ and SGS $\langle \bar{q}^{SGS} \rangle$ vertical heat fluxes, respectively.

3 Numerical Setup and Diagnostics

3.1 Stability Conditions

Two distinct thermal stability conditions are considered in the study: moderately stable, selected from the GABLS-1 experiment (Beare et al. 2006; Gadde and Stevens 2021a), and conventionally-neutral conditions (Stieren and Stevens 2022; Hodgson et al. 2023; Lanzilao and Meyers 2024). Table 3 provides a summary of the parameters used in each stability condition, including horizontal components of the geostrophic wind speed (U_{G1} , U_{G2}); Coriolis parameter f_c ; surface roughness height z_o ; initial surface potential temperature θ_{ini} ; surface cooling rate C_r ; free-atmosphere lapse rate γ_∞ ; and background reference temperature θ_o . For all simulations, the reference density ρ_o is set to 1.225 kg/m³.

The SBL results are compared with LES data obtained by Gadde and Stevens (2021a), who used the pseudo-spectral LESGO code incorporating the LASDD model and a grid size

Table 3 Summary of the values of geostrophic wind speed (U_{G1} , U_{G2}); Coriolis parameter f_c ; surface roughness z_o ; initial surface potential temperature θ_{ini} ; surface cooling rate C_r ; free-atmosphere lapse rate γ_∞ ; and background reference temperature θ_o assigned in the SBL and CNBL cases

	SBL	CNBL
(U_{G1}, U_{G2}) (m/s)	(8,0)	(9.5,0)
f_c (s^{-1})	0.000139 (72.4° lat.)	0.0001185 (54.3° lat.)
z_o (m)	0.1	0.001
θ_{ini} (K)	265	277
C_r (K/h)	0.25	0.00
γ_∞ (K/m)	0.01	0.001
θ_o (K)	263.5	277
Reference	Gadde and Stevens (2021a)	Hodgson et al. (2023)

of $(\Delta x, \Delta y, \Delta z) = (9, 9, 5)$ m, with $D/\Delta \approx 12$. For the CNBL, results are compared with LES data from Hodgson et al. (2023). Their study employed two LES solvers: a finite-volume solver with the TKE model using an isotropic grid size of $D/\Delta \approx 15$, and a pseudo-spectral code with the LASDD model and grid dimensions of $(\Delta x, \Delta y, \Delta z) = (7.8125, 7.8125, 5.86)$ m, i.e., of $D/\Delta \approx 13$.

3.2 Numerical Setup

For the discretization of the computational domain, 7.5-m isotropic grid cells ($D/\Delta = 12$) are used, where $D = 90$ m is the turbine diameter in this study. This grid, designated as D12, provides 12 cells across the turbine diameter, a resolution widely adopted for actuator-disc applications (Calaf et al. 2010; Gadde and Stevens 2019). Notably, the SBL precursor flow is characterized by low atmospheric turbulence, challenging the grid's ability to resolve small turbulent length scales, in contrast to conventionally-neutral or convective boundary layers as well as wind-farm flows which exhibit higher turbulence (Hodgson et al. 2023; Syed et al. 2023). Thus, for the SBL precursor flow, assessing the D12 grid results relative to finer and coarser grids is necessary. For the finer grid, 2-m isotropic grid cells ($D/\Delta = 45$) are used. This grid, providing 45 cells across the turbine diameter and designated as D45, is specifically selected because it also provides more than 80 grid cells across the boundary-layer height. This aligns with the recommendation by Sullivan et al. (2016), which suggests that at least 60 grid cells across the boundary-layer height are necessary to ensure an appropriate resolution of the flow. The 2-m grid spacing also aligns with resolutions used in previous studies by Gadde et al. (2021a) and Beare et al. (2006). For the coarser grid, 15-m isotropic grid cells ($D/\Delta = 6$) are used. This grid, providing 6 cells across the turbine diameter and designated as D6, does not meet the ADM resolution requirements for the 90-m diameter turbines used in this study. However, it is selected because this grid size would satisfy the ADM requirements for larger turbine diameters (e.g., the IEA-15-240-RWT turbine). Therefore, it is of practical interest to investigate how the different SGS models perform with such grid spacing under challenging stable conditions. In summary, the D12 grid is the primary discretization used for the computational domain, while the D45 and D6 grids are also employed for the discretization of the SBL precursor domain.

The computational domain has dimensions $L_x \times L_y \times L_z = 12,000 \times 4800 \times 3840$ m (corresponding to $133D \times 53D \times 43D$) in the x , y , and z directions, respectively (Fig. 2).

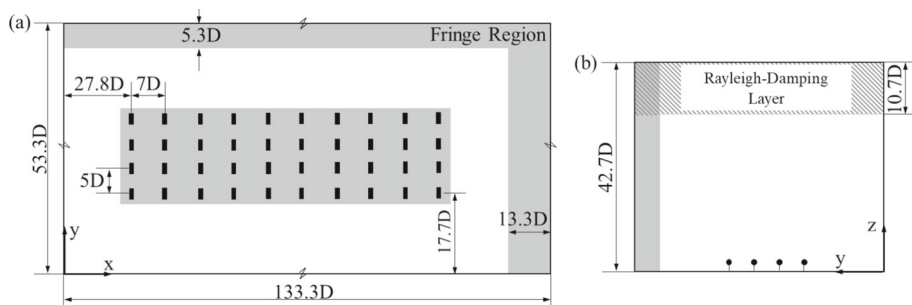


Fig. 2 Schematic showing **a** top view and **b** front view of the computational domain

For the SBL precursor flow, the D6 and D12 grids have also been used for simulations within a smaller domain of $L_x = L_y = L_z = 400$ m (equivalent to $4.5D$), similar to the domain employed by Gadde et al. (2021a) in their simulation of the same SBL flow. This reduction in domain size does not affect the averaged profiles of flow statistics. Therefore, for the SBL precursor flow, the D45 grid simulations are conducted within the smaller domain for computational aspects.

In the SBL precursor and concurrent simulations run with the D12 grid, a fixed time step of 0.5 s is used to ensure Courant–Friedrichs–Lewy (CFL) values remain below 0.6. In the SBL precursor simulations run with the D6 and D12 grids, fixed time steps of 0.5 s and 0.125 s are used, respectively, to maintain the same CFL condition. The CNBL precursor and concurrent simulations are run only using the D12 grid with a time step of 0.375 s to limit CFL values below 0.6. Details of the number of cores and computational times for the different simulations are provided in Appendix C.

In the precursor simulations, doubly periodic boundary conditions are applied in both horizontal directions. At the top boundary, a slip condition is adopted (i.e., the vertical velocity, stresses, and heat fluxes are set to zero). The bottom boundary condition is a rough-wall function with surface-roughness height z_o and surface cooling C_r (Table 3). The precursor boundary-layer flow is initialized with uniform velocity equal to the geostrophic wind. For the SBL, the initial temperature profile is constant over the first 100 m, above which a free-atmosphere lapse rate is applied. For the CNBL, the temperature profile is initialized linearly with the free-atmosphere lapse rate. Uniformly distributed random perturbations with an amplitude of 3% of the geostrophic wind are initially added to velocities below a height of 50 m to spin up turbulence. The mean wind-angle controller (Appendix A.1) is used in the precursor simulation to rotate the entire flow field such that the mean wind direction is aligned with the wind farm (x -axis) at hub height, and which is turned off after the quasi-steady state is reached (i.e., after t_{SS}). Precursor simulations are run using each SGS model until a quasi-steady state is achieved. The SBL reaches the quasi-steady state after 8 h, while the CNBL requires 25 h to achieve a quasi-steady state. After achieving the quasi-steady state, the precursor simulations of both cases are run for two additional hours used to feed the concurrent simulations. Flow statistics are calculated over the last simulation hour (10th hour for the SBL and 27th hour for the CNBL).

In the concurrent simulations, a wind farm comprising ten rows of four turbines is introduced as shown in Fig. 2a. The turbines are modeled using an actuator-disc method with hub height $z_h = 90$ m, diameter $D = 90$ m, and modified thrust coefficient $C'_T = 1.33$ (Eqs. (31) and (33) in Appendix A.4). Turbines have an active local yaw controller to prevent any local onset flow misalignment (Appendix A.4), which may arise from variations in

local wind direction throughout the wind farm due to flow deceleration and Coriolis effects (Allaerts 2016). The concurrent simulation of each SGS model is started from its precursor simulation at t_{SS} (i.e., 8th hour for the SBL and 25th hour for the CNBL), and runs for two hours. The first hour (approximately three flow-through times) is dedicated to solution development, followed by the collection of flow statistics during the second hour. The boundary conditions employed in the precursor simulations are retained in the concurrent simulations. Additionally, a Rayleigh-damping layer, with length $L_z^{rd} = 960 \text{ m}$ ($10.7D$) and damping coefficient $\tilde{\nu} = 0.016 \text{ s}^{-1}$, is added at the top 25% of the domain to prevent instabilities caused by reflected gravity waves triggered by the farm. Two fringe regions are adopted in the concurrent domain (Fig. 2). One fringe region extends in the x direction over the last 1.2 km ($13.3D$) of the domain length (10% L_x), and another extends in the y direction over the last 0.48 km ($5.3D$) of the domain width (10% L_y).

3.3 Diagnostics

The impact of the SGS model on the flow field is quantified through a series of diagnostics. Initially, we provide horizontally and temporally averaged vertical profiles of key variables. For boundary-layer flow, we provide vertical profiles of horizontal wind speed $\langle \tilde{V} \rangle = (\langle \tilde{u}^2 \rangle + \langle \tilde{v}^2 \rangle)^{0.5}$; wind direction $\langle \tilde{\phi} \rangle$; and potential temperature $\langle \tilde{\theta} \rangle$. For turbulence quantities we provide vertical profiles of streamwise turbulence intensity $\langle \overline{TI}_u \rangle = \sqrt{\langle \overline{u'u'} \rangle} / U_\infty$, where $U_\infty = \langle \tilde{V}(z_h) \rangle$ is the horizontal and time-averaged hub-height wind speed; total shear stress $\langle \overline{\tau} \rangle$; SGS dissipation $\langle \tilde{\varepsilon}^{SGS} \rangle = \langle \tau_{ij}^{SGS} \tilde{S}_{ij} \rangle$, normalized by the shear production $\langle \overline{P}_K \rangle = \langle \tau_{ij} \tilde{S}_{ij} \rangle$; and total vertical heat flux $\langle \overline{q} \rangle$. These profiles facilitate direct comparison with similar profiles from existing literature. Additionally, integral output variables characterizing the ABL flow can be derived from these profiles, including jet speed V_{jet} (wind speed maximum value), normalized by the geostrophic wind speed $V_G = (U_{G1}^2 + U_{G2}^2)^{0.5}$; jet height z_{jet} (vertical elevation at which V_{jet} is reached); geostrophic wind direction $\phi_G = \langle \tilde{\phi}(L_z) \rangle$, where $\tilde{\phi} = \arctan(\tilde{v}/\tilde{u})$; inversion height z_c (vertical elevation at which the gradient of the mean temperature profile is maximum); streamwise turbulence intensity at hub height $\langle \overline{TI}_u(z_h) \rangle$; boundary-layer height z_i ; friction velocity u_* , where $u_* = \sqrt{\langle \overline{\tau}_w \rangle}$; surface heat flux $\langle \overline{q}_w \rangle$; and Obukhov length $L_O = -\theta_o u_*^3 / (\kappa g \langle \overline{q}_w \rangle)$. Furthermore, the efficiency of these boundary-layer flow simulations is assessed through the ratio of resolved to total turbulent kinetic energy ρ_{TKE} . For wind-farm simulations, the impact of the farm on the mean ABL flow and its variation across different SGS models is investigated by providing profiles that show changes in the mean flow field relative to the precursor flow. The deviation from the precursor flow is denoted by $\delta(\cdot) = \langle \cdot \rangle_F - \langle \cdot \rangle_{pre}$, where $\langle \cdot \rangle_F$ represents averaging over the horizontal extent of the wind farm (plan area $70D \times 20D$ in Fig. 2a) and $\langle \cdot \rangle_{pre}$ indicates values from the precursor simulation. To elucidate the relative performance of the different SGS models, we also provide the resolved and modeled components of shear stresses and heat fluxes, with $\cdot^{RES}$ and $\cdot^{SGS}$ denoting the resolved and SGS components, respectively.

After the analysis of vertical profiles, we present contours of time-averaged horizontal velocity at both top-tip height and hub height. These contours help in qualitatively assessing the interaction between the wind farm and the ABL flow, as well as the wakes generated

by the turbines. This is followed by spanwise profiles at hub height at downstream distances of $4D$ downstream of the first row, and $4D$, $10D$, and $20D$ downstream of the last row. After examining the wakes of individual turbines, we further investigate wake development both within and downstream of the wind farm through streamwise distributions of time-averaged normalized streamwise velocity $\langle \tilde{u}(z_h) \rangle_{WF,N}$; and streamwise turbulence intensity $\langle \overline{TI}_u(z_h) \rangle_{WF}$ at hub height. Here, $\cdot_N$ denotes normalization by the inflow value, and $\langle \cdot \rangle_{WF}$ indicates averaging in the spanwise direction over the wind farm width of $20D$ (Fig. 2a). To evaluate farm-induced turbulence, we monitor the hub-height streamwise velocity $\langle \tilde{u}(z_h) \rangle_{WF}$; and normalized streamwise turbulence intensity $\langle \overline{TI}_u(z_h) \rangle_{WF,N}$. Given the substantial impact of the supergeostrophic jet on wake recovery and farm performance (Gadde and Stevens 2021a,b; Gadde et al. 2021b), we also examine the streamwise distributions of time-averaged normalized jet speed $\langle \tilde{V}_{jet} \rangle_{WF,N}$; and streamwise turbulence intensity at jet height $\langle \overline{TI}_u(z_{jet}) \rangle_{WF}$, along with an integral measure indicating the percentage reduction in jet strength over the farm area, expressed as $1 - \langle \tilde{V}(z_{jet}) \rangle_F / \langle \tilde{V}(z_{jet}) \rangle_{pre}$. The farm's impact on flow below the turbine rotors is quantified through streamwise distributions of normalized streamwise velocity and streamwise turbulence intensity at the first grid cell off the wall. The vertical turbulent transport of mean kinetic energy across the wind-farm bottom boundary is also quantified as $\langle \tilde{u} \cdot \overline{u'w'}(z_h - R) \rangle_{WF,N}$. The total farm power production P_F is used as an estimate of overall farm performance and is calculated as the sum of the row-averaged power contributions from individual rows (Appendix D). Finally, an energy budget analysis over the different farm rows (Appendix D) is conducted for each simulation to understand the phenomena governing power production.

4 Results

4.1 Stable Boundary Layer

4.1.1 Precursor Boundary-Layer Flow

For the SBL flow, the horizontal and time-averaged vertical profiles, computed over the 10th hour, are presented in Fig. 3 using three grid sizes (D6, D12, and D45). These results are compared with those of Gadde and Stevens (2021a), obtained using a higher-order pseudo-spectral code with a grid size of $D/\Delta \approx 12$. Across all simulations, the jet speed V_{jet} is approximately $1.18V_G \pm 3.5\%$, exhibiting low sensitivity to both the SGS model and grid size (Fig. 3a–c), and closely matching the value reported by Gadde and Stevens (2021a). Conversely, the jet height shows a dependency on the SGS model and grid size. For the D6 grid (Fig. 3a), the WALE, TKE, and LASDD models predict higher normalized jet heights compared to the Smag., AMD, and SDS models. For the D12 grid (Fig. 3b), the TKE model aligns closely with the Smag., AMD, and SDS models, with these four models predicting an average normalized jet height of $z_{jet}/z_h = 1.76$. However, the WALE and LASDD models continue to predict higher jet heights. For both D6 and D12 grids, the larger jet heights predicted by the WALE and LASDD models result in lower wind speeds across the vertical extent of the turbine rotors. With the D45 grid (Fig. 3c), all models exhibit nearly identical wind speed profiles, predicting a normalized jet height of 1.80, similar to that reported by

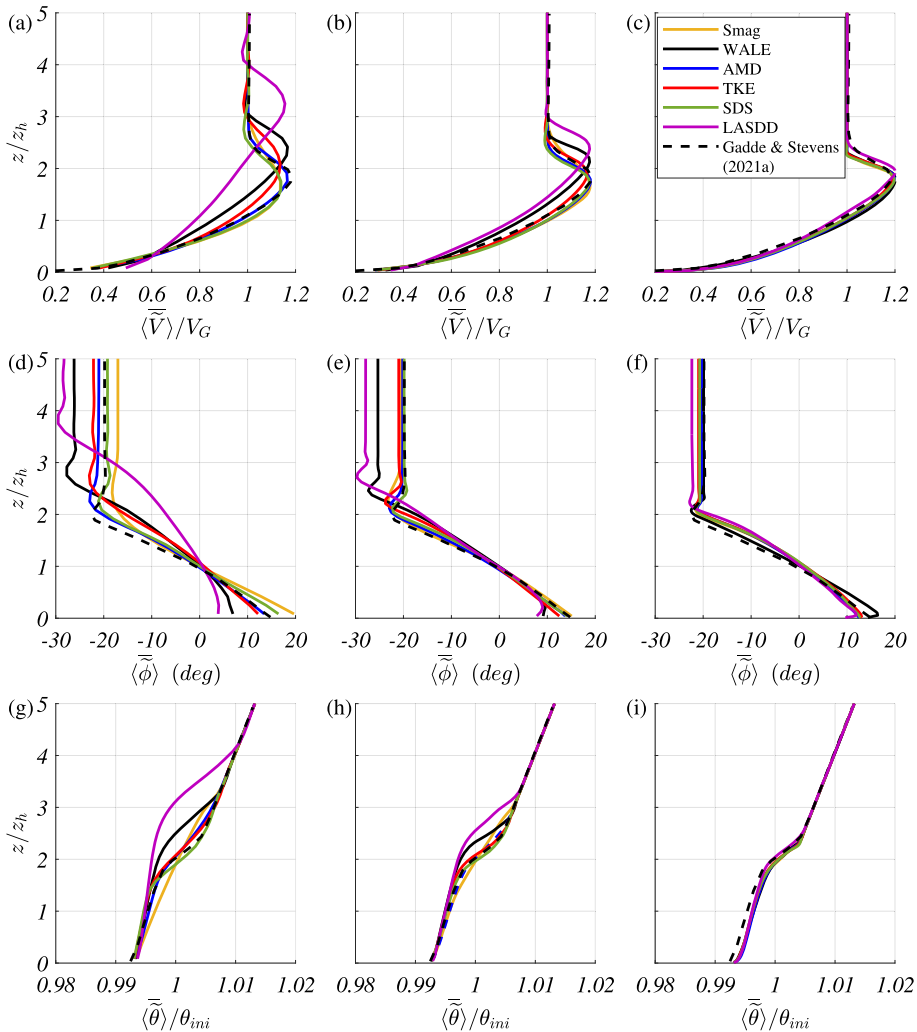


Fig. 3 Vertical profiles of horizontal and time-averaged **a–c** normalized wind speed, **d–f** wind direction, and **g–i** normalized potential temperature for D6 (left), D12 (middle), and D45 (right) grids, obtained from the precursor simulation for the SBL case with the different SGS models. Results from Gadde and Stevens (2021a), obtained with a higher-order pseudo-spectral LES code, are added for comparison. Vertical coordinates are normalized by turbine hub height z_h

Gadde and Stevens (2021a), except for the LASDD model, which predicts an 8% higher value.

The trend exhibited by the different models in predicting the jet height z_{jet} consistently aligns with their corresponding predictions of the boundary-layer height z_i (Figs. 3a–c, 4d–f, and Table 10) and inversion height z_c (Fig. 3g–i and Table 9), except for the Smag. model, which fails to capture the inversion layer using the D6 and D12 grids (Fig. 3g, h). For the D45 grid, all models predict normalized inversion and boundary-layer heights of 2.10 and 1.79, respectively, except the LASDD model, which predicts a 12.3% higher boundary-layer. These values are in very good agreement with the LES results from Gadde and Stevens

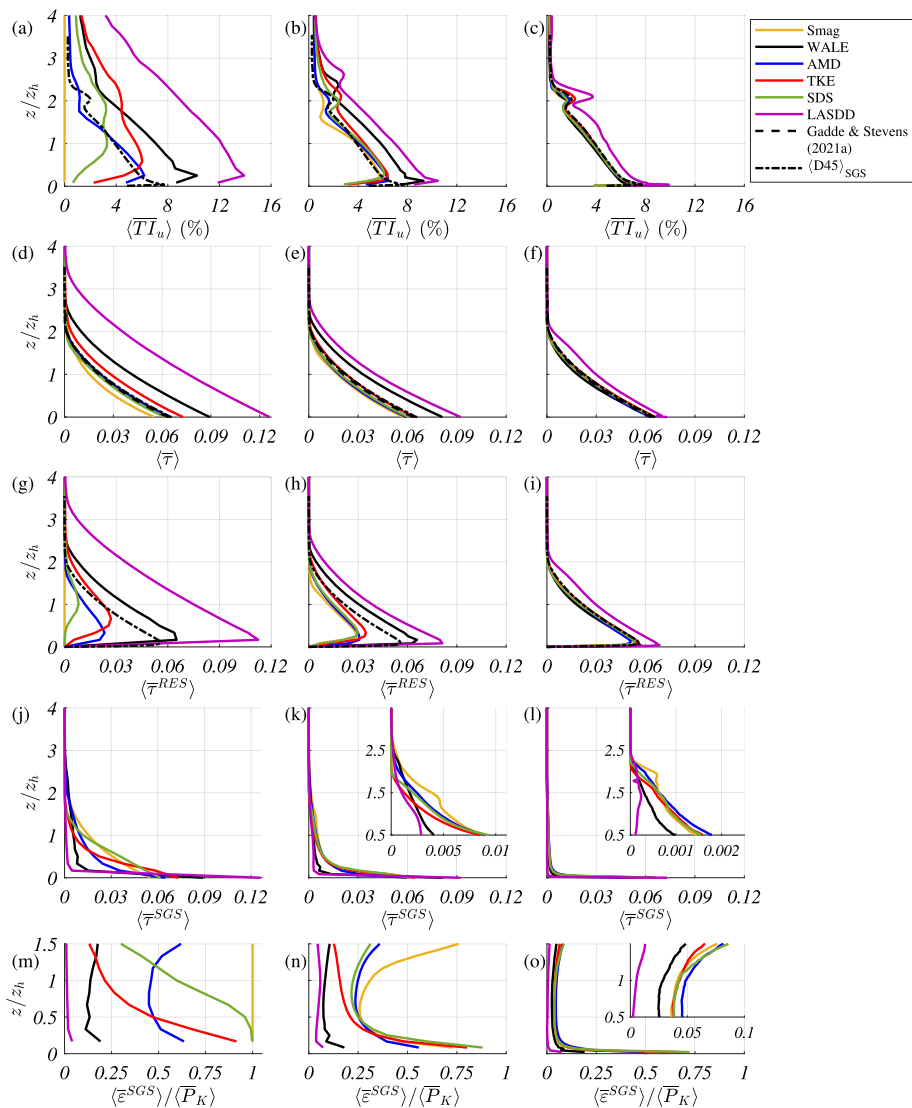


Fig. 4 Vertical profiles of horizontal and time-averaged **a–c** streamwise turbulence intensity, **d–f** total shear stress, **g–i** resolved shear stress, **j–l** SGS shear stress, and **m–o** SGS dissipation normalized by the shear production, obtained with the D6 (left), D12 (middle), and D45 (right) grids from the precursor simulation for the SBL case using the different SGS models. Results from Gadde and Stevens (2021a) are added for comparison, together with a reference profile, denoted by $(D45)_{SGS}$, obtained by ensemble-averaging the profiles from all six SGS models with the D45 grid

(2021a). Moreover, a discussion on the heat fluxes that contribute to the variations in the potential temperature profiles between the models is provided in Appendix C.

The wind direction profiles obtained with the D6 grid (Fig. 3d) are generally consistent across all simulations, with the exception of the WALE and LASDD models, which predict higher values of geostrophic wind direction. Furthermore, the WALE and LASDD models

deviate from the linear wind direction variation over the turbine rotor height predicted by the other four models. The results obtained with the D12 grid (Fig. 3e) show less disparity in wind direction over the turbine's rotor, however, values of ϕ_G are the same as those obtained using the D6 grid. For the D45 grid (Fig. 3f), the vertical profiles of mean wind direction are similar across all simulations with $\phi_G \approx -20^\circ$, except for the LASDD model, which predicts $\phi_G = -22.3^\circ$.

For the turbulence quantities (Fig. 4), discrepancies among the various SGS models are observed in the streamwise turbulence intensity $\langle \overline{Tl_u} \rangle$ profiles for the D6 grid (Fig. 4a). Using the D6 grid, only the AMD model closely matches the ensemble-averaged profile (i.e., the average of profiles from all six SGS models using the D45 grid, denoted by $\langle D45 \rangle_{SGS}$). The D12 grid results (Fig. 4b) show general agreement with the ensemble-averaged D45 grid data, except for the WALE and LASDD models, which predict higher turbulence intensities at hub height. For the D45 grid, all models consistently predict the same turbulence intensity distribution, except for the LASDD model, which predicts a higher hub-height value of 4.9% (Fig. 4c). The trend exhibited by the WALE and LASDD models in predicting the turbulence intensity aligns with their predictions of higher shear stresses.

For the total shear stress, the WALE and LASDD exhibit higher total shear stress compared to the other models when using the D6 grid (Fig. 4d). The total stress profiles obtained with the AMD and SDS models at this coarse grid size match those reported by Gadde and Stevens (2021a). The D12 grid results (Fig. 4e) indicate a reduced disparity in total shear stress among the different simulations, compared to the D6 grid results. For the D45 grid (Fig. 4f), all models present similar total shear stress profiles that match the LES results from Gadde and Stevens (2021a), with ensemble-averaged values of $\tau_w = 0.0659 \text{ m}^2/\text{s}^2$. However, LASDD predicts an 11% higher wall shear stress compared to the ensemble-averaged value. The contribution of the SGS model in predicting the momentum fluxes can be quantified by decomposing the total stress into its resolved and SGS components.

The sensitivity of the resolved and SGS stresses to both the grid size and SGS model is assessed relative to the D45 ensemble-averaged profiles, denoted by $\langle D45 \rangle_{SGS}$. The ensemble-averaged total stress profile matches the results from Gadde and Stevens (2021a). The choice of SGS model influences both the resolved and modeled shear stresses for the D6 grid (Fig. 4g, j), with SGS stresses contributing to the total shear stress up to altitudes reaching twice the hub height. With the Smag. model, nearly all eddies and fluxes are modeled, rather than resolved, using the D6 grid. For the D12 grid (Fig. 4h, k), the disparity in resolved stress among the SGS models decreases, with SGS stresses being predominant only near the bottom wall and becoming negligible above the hub height. Both the WALE and LASDD models predict higher resolved stress relative to the ensemble-averaged profile obtained using the finer grid. The impact of the SGS model choice on the resolved shear stress becomes negligible for the finer D45 grid, as most of the eddies and turbulent fluxes in the flow are resolved (Fig. 4i, l). However, the LASDD model still exhibits slightly higher shear stress values over $0 \leq z/z_h \leq 2$ relative to the other SGS models. The variations exhibited across the different models in predicting the resolved turbulence intensity and shear stresses can be linked to their SGS dissipation.

Vertical profiles of the SGS dissipation $\langle \overline{\epsilon}^{SGS} \rangle$ relative to shear production $\langle \overline{P_K} \rangle$ provide insight to the differing performance of the SGS models, especially in the near-wall region, where these terms dominate the turbulent kinetic energy budget (Pope 2000; Pedersen et al. 2014). With the D6 grid, the Smag. model exhibits the highest dissipation relative to other SGS models (Fig. 4m), achieving a full equilibrium between dissipation and shear production. This equilibrium indicates that the grid fails to resolve any turbulent kinetic energy, a limitation of the Smag. model when the filter width associated with the grid size exceeds the turbulence

Table 4 Turbulent kinetic energy ratio ρ_{TKE} obtained with the grids D6, D12, and D45 using the different SGS models

Grid	ρ_{TKE} (%)					
	Smag.	WALE	AMD	TKE	SDS	LASDD
D6	0.0	93.5	80.3	81.1	48.9	99.9
D12	77.7	92.8	85.7	81.4	79.8	98.7
D45	91.2	93.05	91.6	88.65	89.73	99.8

length scales (Pope 2000). In contrast, the WALE and LASDD models show the lowest dissipation relative to shear production. As the grid becomes finer, smaller eddies are resolved, and the contribution of the SGS model to the energy cascade decreases, except near the wall where the grid resolution is insufficient to resolve near-wall eddies. For the D12 grid, normalized dissipation values are lower compared to the D6 grid beyond $z/z_h > 0.5$ (Fig. 4n). With this grid size, the normalized dissipation from the Smag. model is similar to those of the AMD, TKE, and SDS models, while lower dissipation rates persist for WALE and LASDD. For the D45 grid (Fig. 4o), SGS dissipation is significant only near the bottom wall. All simulations produce similar profiles of normalized dissipation, except for the LASDD model, which shows slightly lower values. These findings for the LASDD SGS model correlate with the higher turbulence intensity values (Fig. 4c).

After the discussion of the vertical profiles as obtained by the precursor simulations, we further explore the relative efficiency of the simulations through the turbulent kinetic energy ratio ρ_{TKE} . This ratio represents the extent to which the grid size allows for the turbulent kinetic energy to be resolved (Table 4). Although ρ_{TKE} cannot be considered a definitive measure of the accuracy of LES of stable boundary-layer flows (Sect. 5), it is widely used as an indicator of LES efficiency (Pope 2000). As the grid is refined, a larger proportion of the turbulent kinetic energy is resolved, while a smaller proportion is modeled by the SGS model. This trend is consistent across all SGS models, except for the WALE and LASDD models, which maintain nearly constant ρ_{TKE} values above 90% across all three grid sizes, attributed to their low SGS dissipation (Fig. 4m–o). For all grids, the AMD and TKE models resolve more than 80% of the turbulent kinetic energy, thereby meeting a commonly accepted threshold for LES effectiveness (Pope 2000). The Smag. and SDS models only achieve this threshold with the finest D45 grid. Finally, Appendix C summarizes the key integral metrics obtained for the three grid sizes and six SGS models, including normalized jet speed, normalized jet height, geostrophic wind direction, normalized inversion height, normalized friction velocity, normalized boundary-layer height, hub-height streamwise turbulence intensity, surface heat flux, and normalized Obukhov length.

The challenging SBL precursor simulations are conducted using three different grids to enhance the understanding of the D12 grid results, in comparison to other LES studies employing high-order pseudo-spectral codes (Gadde and Stevens 2021a), as well as the finer D45 grid results. Results from the SBL flow using the D45 grid are largely insensitive to the SGS model. However, this grid size is computationally expensive, requiring nearly 28 billion grid cells to uniformly discretize the domain for the 40-turbine wind-farm simulation. The D12 grid results show some sensitivity to the SGS model but largely agree with those from the D45 grid (Appendix C) and with findings from Gadde and Stevens (2021a), while incurring substantially lower computational costs (524 million grid cells). Additionally, the D12 grid offers 12 grid cells across the turbine diameter, a widely used resolution with the actuator-disc method (Calaf et al. 2010; Gadde and Stevens 2019). The coarse D6 grid shows large deviations from the fine grid results and differences between SGS models and is therefore

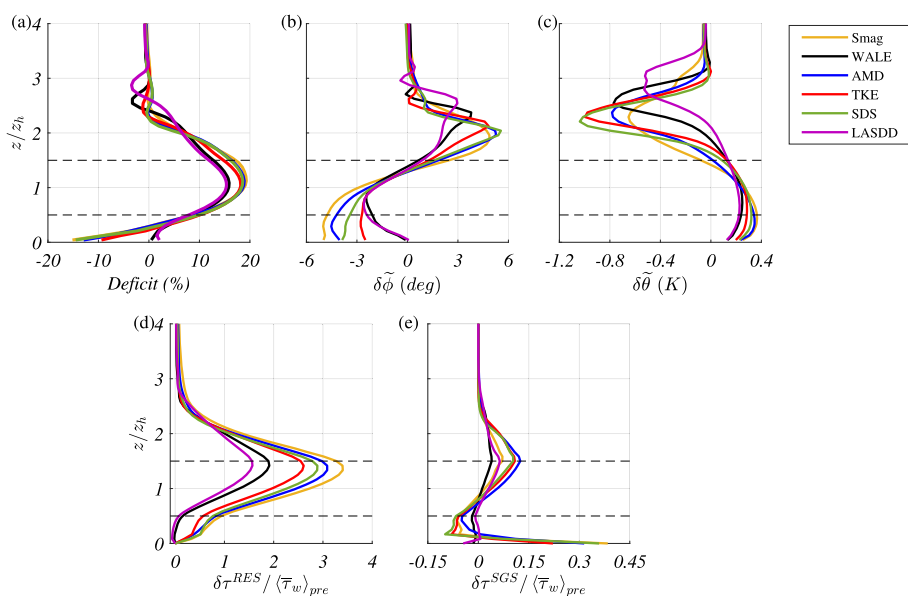


Fig. 5 Horizontal (over the $70D \times 20D$ region surrounding the wind farm) and time-averaged profiles from the concurrent simulation of the SBL case, obtained with the SGS models adopting the grid D12, of **a** velocity deficit, **b** change in wind direction, **c** change in temperature, **d** normalized change in resolved shear stress, and **e** normalized change in SGS shear stress. Dashed lines indicate top and bottom rotor tip heights

deemed inadequate for adoption in the current wind-farm simulations. Consequently, the D12 grid is adopted for running the concurrent wind-farm SBL simulations and CNBL precursor and concurrent simulations.

4.1.2 Wind-Farm Flow

This section presents the results of wind-farm flow simulations conducted under stable conditions using the six SGS models and the D12 grid. We start by quantifying the impact of the wind farm on the mean wind speed through vertical profiles of the velocity deficit (Fig. 5a). The Smag. model predicts a maximum velocity deficit of 19.3%, while the AMD, TKE, and SDS models yield maximum deficits of 18.6%. The WALE and LASDD models, on the other hand, predict lower maxima of 15.9% and 15.4%, respectively. Notably, the Smag., AMD, TKE, and SDS models predict a negative deficit (i.e., an increase in velocity) ranging from -10 to -15% near the bottom wall. In contrast, the WALE and LASDD models show a velocity deficit. This discrepancy is further investigated throughout the remainder of the study. Changes in the mean wind speed are accompanied by changes in the mean wind direction due to Coriolis forcing and vertical turbulent transport of lateral momentum (van der Laan and Sørensen 2017; Howland et al. 2018).

All models demonstrate a clockwise shift in the mean wind direction $\delta\tilde{\phi}$ (i.e., wind moving in the negative y -direction) near the bottom wall when compared to the precursor simulation (Fig. 5b). The maximum clockwise rotation of the flow occurs at the bottom-tip height, with angles ranging from -2.5° to -5° . Across the vertical extent of the turbine rotors, all simulations indicate a diminishing clockwise rotation of the mean wind direction, approaching zero at top-tip height. Above the turbine level ($1.5 < z/z_h < 3$), all simulations consis-

tently exhibit a counterclockwise rotation of the mean wind direction (i.e., in the positive y -direction) relative to the precursor flow, with maximum rotation angles of 3° for the LASDD model and 5.5° for the AMD and SDS models. The operation of the turbines also induces a change in the mean temperature field $\delta\hat{\theta}$ on the order of 1 K relative to the precursor flow (Fig. 5c). All simulations predict a relative increase in the mean potential temperature over the vertical extent of the turbine rotors and below the wind farm. Conversely, above the wind farm, a reduction in potential temperature is consistently predicted, with the maximum decrease occurring near the top of the boundary layer. The Smag. and AMD models predict the highest temperature rise, reaching 0.38 K at the bottom-tip height, whereas the TKE and SDS models predict a temperature drop of -1 K at $z/z_h \approx 2.25$. In comparison, the WALE and LASDD models predict relatively smaller temperature variations of -0.8 K and -0.54 K, respectively (Fig. 5c).

For the turbulence quantities, all simulations predict an increase in resolved shear stress up to $z/z_h \approx 4$, with maxima observed slightly below the top-tip height (Fig. 5d). Although the WALE and LASDD models exhibit relatively higher resolved shear stress over the vertical extent of the turbine rotors in the precursor flow (Fig. 4h), these two models exhibit lower farm-induced shear stress compared to the other SGS models. In contrast, the Smag. and AMD models predict the highest farm-induced resolved shear stress across the vertical extent of the turbine rotors. Additionally, an increase in SGS shear stress is observed over the vertical extent of the turbine rotors, extending up to $z/z_h \approx 2.5$ (Fig. 5e), with the AMD simulation predicting the highest values. Results from the Smag., AMD, TKE, and SDS simulations show an initial increase in SGS stress near the wall, followed by a decrease relative to the precursor flow up to the bottom-tip height ($z = z_h - R$). Conversely, the WALE and LASDD simulations indicate a negligible role for SGS stress below the turbine rotors. After discussing the impact of the wind farm on the mean flow and the sensitivity of this impact to the SGS model, we study the variations across the different SGS models in predicting the turbine wakes.

The contours of the time-averaged horizontal velocity, obtained using the AMD model, show the clockwise rotation of the wakes (Fig. 6), which is more pronounced at top-tip height (Fig. 6a) compared to hub height (Fig. 6b). The rotation of the wakes is attributed to wind veer and Coriolis forcing. These factors also change the wakes' shape, which loses its axisymmetric form, featuring elliptical cross-sections with their major axes rotated relative to the horizontal plane (van der Laan and Sørensen 2017; Howland et al. 2018; Ghobrial et al. 2024). A more quantitative analysis is provided by examining the wake profiles at various streamwise locations along the wind farm to assess their sensitivity to the SGS model (Fig. 7). Downstream of the first row (Fig. 7a), all simulations display similar spanwise profiles of normalized velocity at hub height within the wake region. Further downstream (Fig. 7b–d), lower velocity deficits are observed on the left side of the wind farm, which can be attributed to the wind veer that deflects the wakes clockwise (Fig. 5b). At the downstream end of the wind farm, the LASDD and WALE models demonstrate higher wake recovery rates (Fig. 7b–d), primarily due to their higher inflow turbulence intensities (Fig. 4b).

The streamwise development of the flow, both through and downstream of the wind farm, is quantified by examining the streamwise distributions of velocity and turbulence intensity at the jet height, hub height, and the first grid cell off the wall (Fig. 8). For the streamwise development of the jet speed (Fig. 8a), the WALE and LASDD simulations exhibit lower momentum extraction from the jet by the wind farm, resulting in higher values of normalized jet speed. The AMD and SDS models show almost identical streamwise evolution of $\tilde{V}_{jet}$, whereas the Smag. model presents slightly lower values compared to AMD and SDS. These results align with the jet heights predicted by these models (Fig. 3b), as well as the reductions

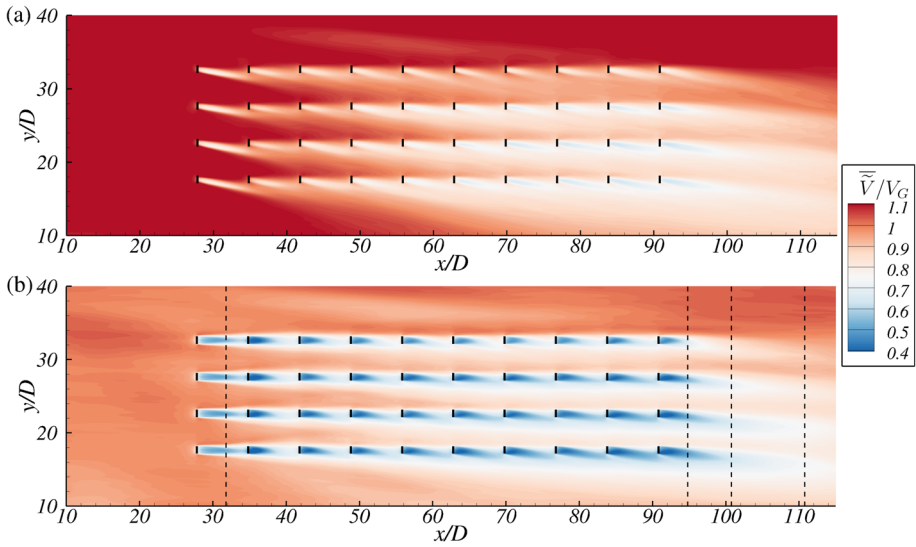


Fig. 6 Horizontal contours of time-averaged horizontal velocity for the SBL case from the AMD simulation at **a** top-tip height ($z/z_h = 1.5$) and **b** hub height ($z = z_h$). Dashed lines indicate locations where the spanwise distributions of velocity are extracted

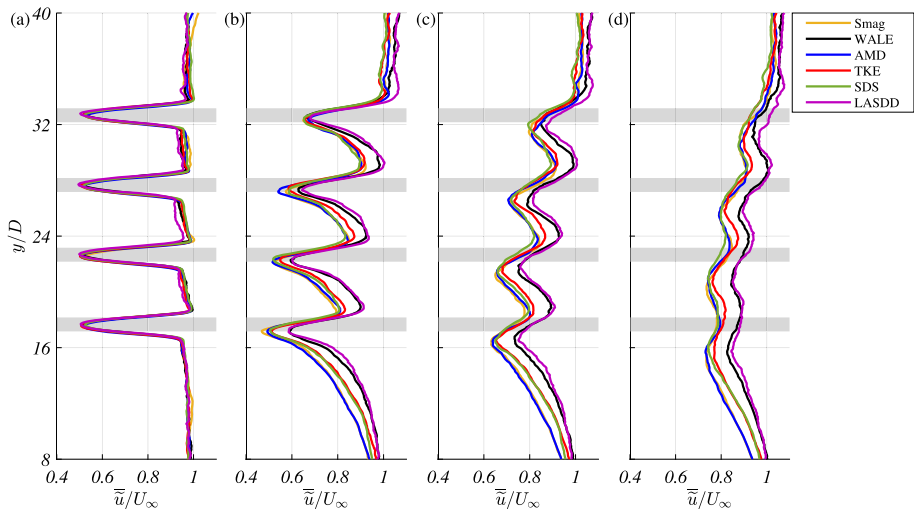


Fig. 7 Spanwise distribution of the time-averaged streamwise velocity $\tilde{u}$ at hub height for the SBL wind-farm flow at a distance: **a** $4D$ downstream of the first row of turbines, **b** $4D$, **c** $10D$, and **d** $20D$ downstream of the last row of turbines (see dashed lines in Fig. 6b). The profiles are normalized by horizontal and time-averaged wind speed obtained from the precursor simulation ($U_\infty = \langle \tilde{V}(z_h) \rangle_{pre}$)

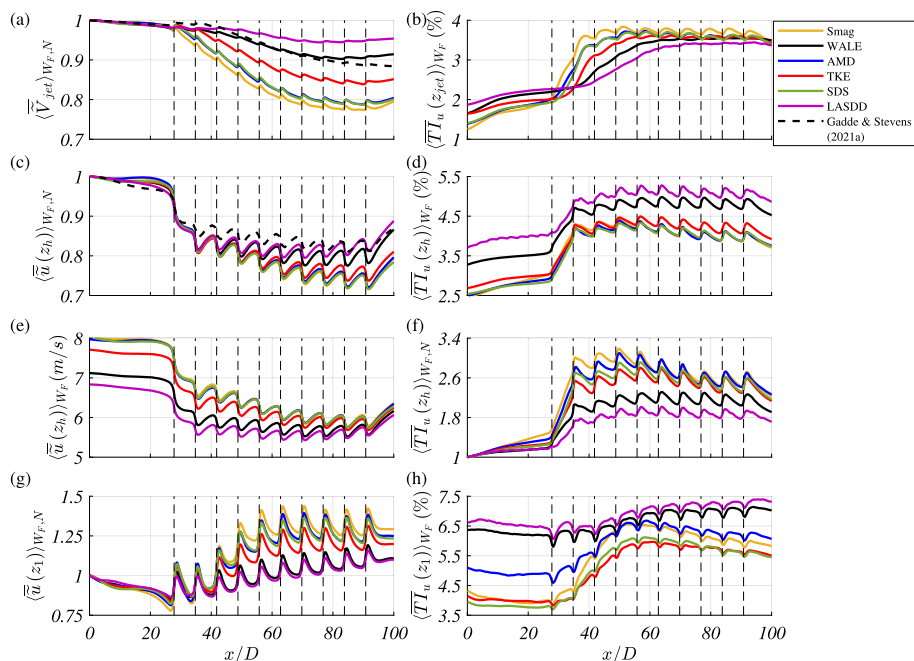


Fig. 8 Streamwise distribution from the SBL case for the spanwise (over the $20D$ region surrounding the farm) and time-averaged **a** normalized jet speed, **b** streamwise turbulence intensity at jet height, **c** and **e** normalized and absolute streamwise velocity at hub height, **d** and **f** absolute and normalized streamwise turbulence intensity at hub height, and **g** and **h** normalized streamwise velocity and absolute streamwise turbulence intensity at the first grid cell off the wall. Results from Gadde and Stevens (2021a), who used a pseudo-spectral LES code, are added for comparison

in jet strength. The Smag., AMD, and SDS models exhibit approximately a 10.0% reduction in V_{jet} , whereas the TKE, WALE, and LASDD models predict lower reductions in jet strength of 7.40%, 4.75%, and 2.75%, respectively. The WALE and LASDD models predict higher inflow turbulence intensity $\overline{TI}_u(z_{jet})$ than the other models; however, after the third row of the wind farm, the Smag., AMD, TKE, and SDS models exhibit higher $\overline{TI}_u(z_{jet})$ due to greater jet–farm interaction (Fig. 8b). Downstream of the farm, all models predict similar streamwise turbulence intensity across the farm’s lateral extent.

At hub height, the Smag., AMD, TKE, and SDS models show similar values of normalized averaged streamwise velocity along the wind farm (Fig. 8c) and exhibit a comparable per-row recovery rate up to the fourth row of turbines. Further downstream, the TKE model predicts a slightly higher wake recovery. The WALE and LASDD models display more substantial differences, particularly near the end of the farm, predicting higher recovery rates. These findings are consistent with the turbulence intensity distributions, where the WALE and LASDD models predict higher values of $\overline{TI}_u(z_h)$ across the entire farm (Fig. 8d). However, when normalized by the inflow turbulence intensity, the WALE and LASDD models predict the lowest farm-induced turbulence (Fig. 8f), whereas the Smag. model predicts the largest farm-induced turbulence. This discrepancy can be partially attributed to the lower inflow wind speed estimated by these SGS models (Fig. 8e), which results in reduced turbine thrust, thereby decreasing the shear and turbulence intensity induced by the wind farm. Additionally,

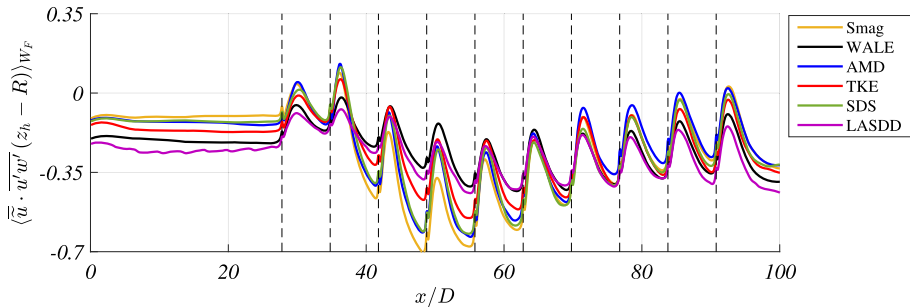


Fig. 9 Streamwise distribution from the SBL case for the spanwise (over the $20D$ region surrounding the farm) and time-averaged vertical turbulent transport of mean kinetic energy across the bottom boundary of the turbine rotors (i.e., at $z = z_h - R$)

in the experiments by Talavera and Shu (2017), it was shown that high inflow turbulence intensity (Fig. 8d) correlates with reduced levels of turbine-induced turbulence.

At the first grid cell off the wall ($z = z_1$), the SGS models vary in predicting the streamwise velocity distribution, primarily due to differences in predicting the jet height. All models show a velocity reduction near the bottom wall compared to the values at the beginning of the computational domain (Fig. 8g). However, the Smag., AMD, TKE, and SDS models predict velocities that exceed those at the inflow after three turbine rows, with maxima observed in the 6th to 7th rows for the Smag. and AMD models. This velocity increase is less pronounced in the WALE and LASDD models, which predict a larger jet height. Regarding turbulence intensity near the wall (Fig. 8h), the WALE and LASDD models predict lower farm-induced turbulence, while the other models predict a higher increase, with values exceeding those at hub height. The vertical turbulent transport of mean kinetic energy across the bottom boundary of the turbine rotors ($z = z_h - R$) is shown in Fig. 9. The Smag., AMD, TKE, and SDS models exhibit higher values compared to the WALE and LASDD models along the first six rows of turbines. These differences are attributed to variations in predicting the altitude of the supergeostrophic jet interacting with the farm (Fig. 3b). The enhanced vertical turbulent transport of mean kinetic energy exhibited by the Smag., AMD, TKE, and SDS models leads to flow acceleration at low altitudes beneath the turbine rotors (Figs. 5a and 8g).

To understand the impact of the SGS model on overall farm performance, we compare the time-averaged farm power production, denoted as P_F , as predicted by the different models. The AMD model predicts P_F to be 25.4 MW, while the Smag. model predicts a 1.18% higher power output. In contrast, the WALE, TKE, SDS, and LASDD models predict reductions in power output of 27.2%, 13.8%, 4.33%, and 34.6%, respectively, when compared to the AMD model (Table 5). Notably, these farm power production estimates are consistent with the velocity deficit profiles (Fig. 5a), as simulations that predict lower power outputs also tend to predict smaller reductions in mean wind speed due to the operation of the wind farm. Moreover, the flow mechanisms governing power production and wake recovery are quantified through the kinetic energy budget (Eq. (54) in Appendix D), evaluated on a per-row basis, i.e., over a control volume encompassing each row (Fig. 20). For the first row of turbines, the WALE, TKE, and LASDD models predict first-row power production $|P|_{row=1}$ values of 3.14 MW, 4.02 MW, and 2.67 MW, respectively, in contrast to the higher value of approximately 4.5 MW predicted by the Smag., AMD, and SDS models (Table 5), which have larger inflow wind speeds at the turbine level (Figs. 3b and 8e). The kinetic energy flux across the west boundary of the wind farm, $|E|_{k,W, row=1}$ (Appendix D), reflects the

Table 5 Results of farm power production P_F ; first-row power production $|P|_{row=1}$; kinetic energy flux $|E|_{k,W row=1}$; and turbulent transport $|T|_{t,W row=1}$ across the west surface of the first row for the SBL wind-farm flow

	Smag.	WALE	AMD	TKE	SDS	LASDD
P_F (MW)	25.7	18.5	25.4	21.9	24.3	16.6
$ P _{row=1}$ (MW)	4.52	3.14	4.53	4.02	4.56	2.67
$ E _{k,W row=1}$ (MW)	48.3	33.6	47.8	42.3	47.6	28.9
$ T _{t,W row=1}$ (MW)	0.248	0.258	0.245	0.230	0.207	0.290

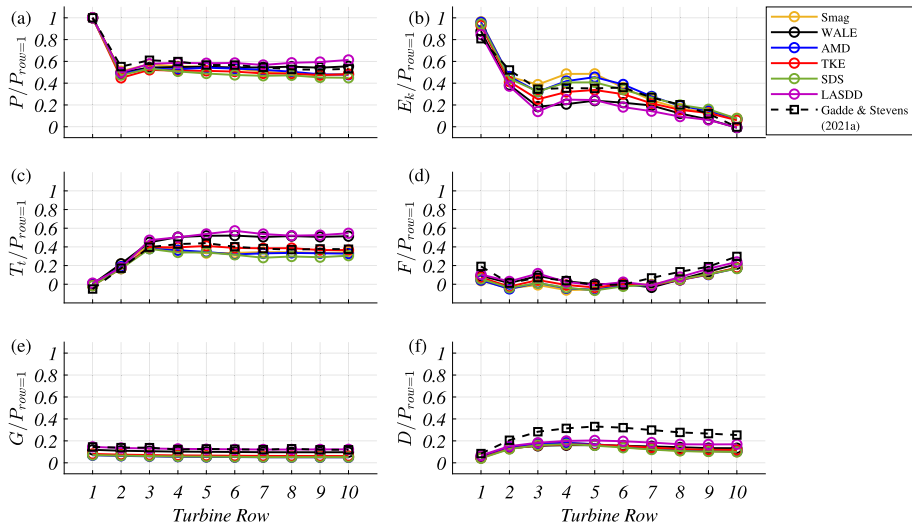


Fig. 10 Row variation of the kinetic energy budget terms (Eq. 54): **a** turbine power production P ; **b** kinetic energy flux E_k ; **c** turbulent transport T_t ; **d** flow work F ; **e** geostrophic forcing G ; and **f** dissipation D , normalized by the power production of the first row $P_{row=1}$, using the different SGS models for the SBL case, compared with LES results from Gadde and Stevens (2021a)

reduced availability of kinetic energy that can be extracted by the first row of turbines. The WALE, TKE, and LASDD models predict $|E|_{k,W row=1}$ as 33.6 MW, 42.3 MW, and 28.9 MW, respectively, compared to an ensemble-averaged value of 47.9 MW estimated by the other models (Table 5). After analyzing the performance of the first row, we use the first-row power production as a normalization value to facilitate comparison with the LES data from Gadde and Stevens (2021a).

For the budget analysis of the different rows, all simulations consistently report a nearly 50% reduction in power production for the secondary rows compared to the first row (Fig. 10a). Downstream of the fourth row, the WALE and LASDD models predict higher normalized power, which aligns with the evolution of the normalized streamwise velocity at hub height (Fig. 8c). As turbines continuously extract energy from the flow, the kinetic energy flux, E_k , decreases with increasing row count (Fig. 10b). After the third row, all models predict a slight increase in E_k , with this increase being more pronounced in the Smag., AMD, TKE, and SDS simulations. The normalized turbulent transport of kinetic energy, T_t (Fig. 10c), is similar across the first three rows for all SGS models. However, the WALE and LASDD models exhibit a larger net normalized T_t in the subsequent rows, while it is lowest

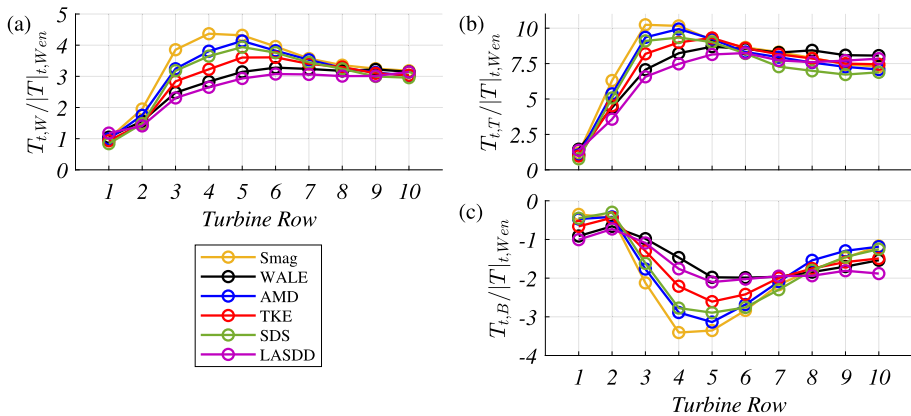


Fig. 11 Row variation of the turbulent transport T_t across the **a** west (W), **b** top (T), and **c** bottom (B) surfaces, normalized by the first row inflow turbulent transport $|T|_{t,Wen}$, ensemble-averaged from the values of the different SGS models (Table 5), for the SBL case. In this figure, positive and negative values indicate transport into and out of the control volume, respectively

in the SDS, AMD, and Smag. models. The row-wise variation of the normalized flow work, F (Fig. 10d), is almost independent of the choice of SGS model. The increase in normalized F observed in the last three rows indicates that these rows operate under favorable pressure gradient conditions, thereby enhancing their power production (Gadde and Stevens 2021a) (Fig. 10a). The normalized geostrophic driving force, G (Fig. 10e), remains nearly constant across all rows, showing limited sensitivity to the SGS model employed. The normalized flow dissipation, D (Fig. 10f), acts as a sink of kinetic energy [as indicated by the negative sign in Eq. (54)], with its value across all rows showing limited sensitivity to the SGS model used, consistently underpredicting dissipation compared to the results from Gadde and Stevens (2021a). The turbulent transport of kinetic energy is a key term in Eq. (54), warranting detailed examination of its fluxes across the surfaces enclosing the control volume.

The primary fluxes involved in the turbulent transport of kinetic energy, T_t , occur across the west, top, and bottom surfaces (Appendix D), with values at the inflow (west) surface, $|T|_{t,W}$, derived from the six SGS models, presented in Table 5. Figure 11 illustrates the per-row evolution of these three fluxes, where positive and negative values indicate transport into and out of the control volume, respectively. The values are normalized by the ensemble-averaged turbulent transport across the west boundary of the first row $|T|_{t,Wen}$. Across the west surface of each row (Fig. 11a), all models exhibit an increase in turbulent transport compared to the first-row value due to turbine-induced turbulence, with the WALE and LASDD models predicting the lowest increase and the Smagorinsky model predicting the highest. Across the top surface of the control volume (Fig. 11b), the Smag., AMD, TKE, and SDS models have the highest downward vertical turbulent transport of kinetic energy into the farm as a result of a lower-altitude supergeostrophic jet (Fig. 3b) and higher turbine-induced turbulence (Fig. 8b, d, and f). These models also show the largest fluxes across the bottom surface, indicating that kinetic energy is transported from the rotors to the region underneath (Fig. 11c). Further downstream, beyond the seventh row, the supergeostrophic jet loses its strength, leading to a reduction in T_t values among the models. Overall, the west fluxes of T_t balance those over the bottom surface, leaving the top surface fluxes to dominate the turbulent transport of kinetic energy across the farm. This emphasizes the importance of accurately simulating the lower-altitude supergeostrophic jet that interacts with the wind-farm rows.

4.2 Conventionally-Neutral Boundary Layer

4.2.1 Precursor Boundary-Layer Flow

The conventionally-neutral boundary layer represents offshore conditions, where surface heat fluxes are substantially lower than those observed onshore (Allaerts 2016). For the CNBL simulations, where only the D12 grid is used, the averaged vertical profiles exhibit less sensitivity to the SGS model (Fig. 12). All six SGS models predict a maximum wind speed of $1.08V_G$ at a normalized jet height of $z_{jet}/z_h = 5.5$ (Fig. 12a), with the exception of the LASDD model, which predicts a jet height that is 4% lower. Additionally, all models predict a capping inversion and boundary-layer heights of $6.48z_h$ and $6.06z_h$, respectively, except for the LASDD model, which predicts a boundary-layer height that is 7.1% lower. These results are consistent with the wind speed profile obtained using a pseudo-spectral LES code (denoted as “EPFL”) from Hodgson et al. (2023) with the LASDD model and a grid size of $D/\Delta \approx 13$. Conversely, results from a finite-volume LES code (denoted as “DTU”) using the TKE SGS model and a grid size of $D/\Delta \approx 15$, also from Hodgson et al. (2023), predict larger jet heights than the present LES data. Generally, weaker wind veer is observed in the CNBL case (Fig. 12b) compared to the SBL case (Fig. 3e). The Smag., AMD, TKE, and SDS models predict a geostrophic wind direction of -12.5° , whereas the WALE and LASDD models predict values of -14.4° and -15.2° , respectively (Fig. 12b). In all simulations, the mean wind direction varies linearly over the vertical extent of the turbine rotors, except for the LASDD model. All models consistently predict the potential temperature distribution (Fig. 12c), which is in excellent agreement with the pseudo-spectral LES code results, while the finite-volume solver from Hodgson et al. (2023) predicts a higher capping inversion height.

For the turbulence quantities, higher turbulence intensities are predicted by the WALE and LASDD models (Fig. 12d), with hub-height values of 6.18% and 7.50%, respectively, whereas the Smag., AMD, TKE, and SDS models predict $\overline{TI}_u(z_h)$ values of 5.53%. In terms of wall-shear stress, the Smag., AMD, TKE, and SDS models predict a value of $0.0812 \text{ m}^2/\text{s}^2$, whereas the WALE and LASDD models predict wall-shear stress values of $0.0879 \text{ m}^2/\text{s}^2$ and $0.0907 \text{ m}^2/\text{s}^2$, respectively. Although the WALE and LASDD models predict relatively higher SGS stresses near the wall, these two simulations exhibit lower SGS stresses compared to the other models throughout the boundary layer. The SGS stresses are dominant only near the wall and decrease with increasing height (Fig. 12e, f). The normalized SGS dissipation profiles (Fig. 12g) show largest values over $z/z_h \leq 0.25$, above which it decreases to values less than 10% of the shear production. The disparity among the SGS models is most pronounced near the wall, with the TKE and SDS models predicting the highest normalized SGS dissipation. In contrast, the WALE and LASDD models provide the lowest normalized SGS dissipation, associated with higher resolved turbulence profiles, as found in the SBL case.

The integral output flow variables obtained for the various SGS models are listed in Table 6. The values demonstrate the reduced sensitivity of the CNBL precursor flow to the choice of SGS model, as noted in previous LES studies (Peña 2019; Doubrawa et al. 2020; Stoll et al. 2020; Hodgson et al. 2023). This reduced sensitivity suggests that the D12 grid is effective in resolving the characteristic eddies with relatively larger length scales typical of the CNBL case, as opposed to the smaller length scales in SBL flows.

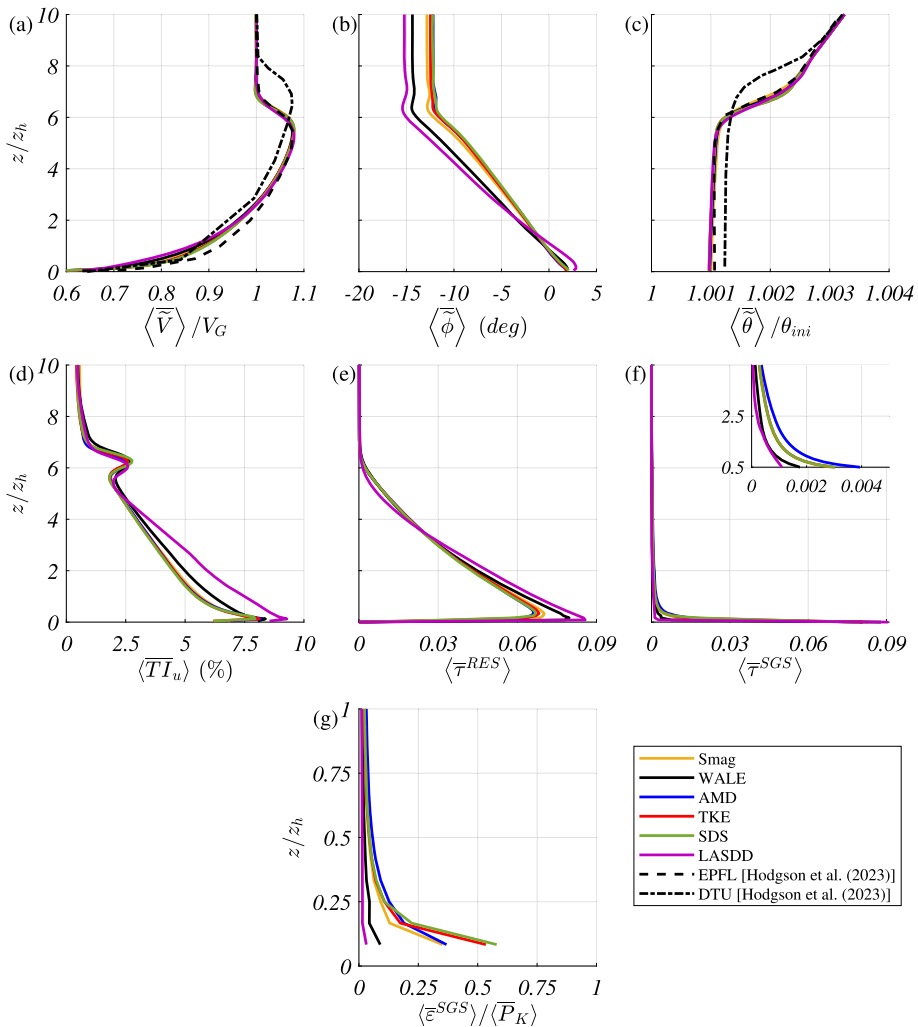


Fig. 12 Horizontal and time-averaged vertical profiles of **a** normalized wind speed, **b** wind direction, **c** normalized potential temperature, **d** streamwise turbulence intensity, **e** resolved shear stress, **f** SGS stress, and **g** SGS dissipation normalized by the shear production, for the precursor CNBL simulation using the D12 grid. Results from finite-volume (“DTU”) and pseudo-spectral (“EPFL”) codes by Hodgson et al. (2023) are added for comparison

4.2.2 Wind-Farm Flow

This section presents the results of wind-farm flow simulations for the CNBL case using the six SGS models and the D12 grid. We start by analyzing the impact of the wind farm on the mean atmospheric flow (Fig. 13). This impact extends from the bottom wall up to an elevation of $z/z_h = 4$, as shown in the velocity deficit profiles (Fig. 13a). All models exhibit an average maximum velocity deficit of 12.9%. Below the turbine rotors, all simulations predict a velocity deficit, contrary to the SBL case, where all models, except for LASDD and WALE, predicted flow acceleration. Wind-farm-induced alterations in wind direction are

Table 6 Results obtained with the D12 grid for the precursor CNBL of normalized jet speed V_{jet}/V_G ; normalized jet height z_{jet}/z_h ; geostrophic wind direction ϕ_G ; normalized inversion height z_c/z_h ; hub-height streamwise turbulence intensity $\langle \overline{T}I_u(z_h) \rangle$; normalized boundary-layer height z_i/z_h ; normalized friction velocity u_*/V_G ; and resolved-to-total turbulent kinetic energy ratio ρ_{TKE}

Model	V_{jet}/V_G	z_{jet}/z_h	$\phi_G (^\circ)$	z_c/z_h	$\langle \overline{T}I_u(z_h) \rangle (\%)$	z_i/z_h	u_*/V_G	$\rho_{TKE} (\%)$
Smag.	1.08	5.46	-12.9	6.71	5.63	6.06	0.0303	93.6
WALE	1.08	5.46	-14.4	6.54	6.18	6.04	0.0312	96.7
AMD	1.08	5.46	-12.2	6.46	5.53	6.07	0.0299	93.5
TKE	1.08	5.54	-12.5	6.46	5.49	6.07	0.0300	91.0
SDS	1.08	5.54	-12.3	6.46	5.46	6.08	0.0297	92.3
LASDD	1.08	5.29	-15.3	6.38	7.50	5.63	0.0317	99.7

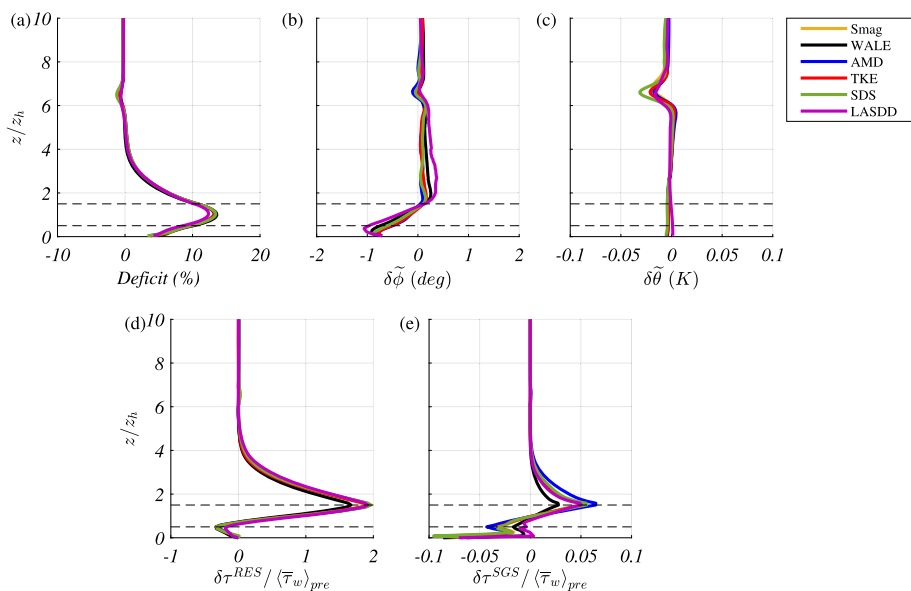


Fig. 13 Horizontal (over the $70D \times 20D$ region surrounding the wind farm) and time-averaged profiles from the concurrent simulation of the CNBL case of **a** velocity deficit, **b** change in wind direction, **c** change in temperature, **d** normalized change in resolved shear stress, and **e** normalized change in SGS shear stress. Dashed lines indicate top and bottom rotor tip heights

minimal, with changes of less than 1° (Fig. 13b). Similarly, variations in mean temperature are negligible, with a slight decrease of approximately 0.02 K observed at the inversion height (Fig. 13c). These variations in the temperature and velocity fields at the inversion height have been reported in other studies of LES of wind farms operating under conventionally-neutral conditions (Allaerts and Meyers 2017; Lanzilao and Meyers 2024). Resolved shear stress increases over the vertical extent of the turbine rotors, extending up to an elevation of $z/z_h \approx 4$, with peak values occurring at top-tip height (Fig. 13d). Regarding the SGS stresses (Fig. 13e), wind-farm-induced variations are considerably smaller than those of the resolved stresses, reflecting the relatively minor contribution of SGS stresses in the CNBL case (Fig. 12).

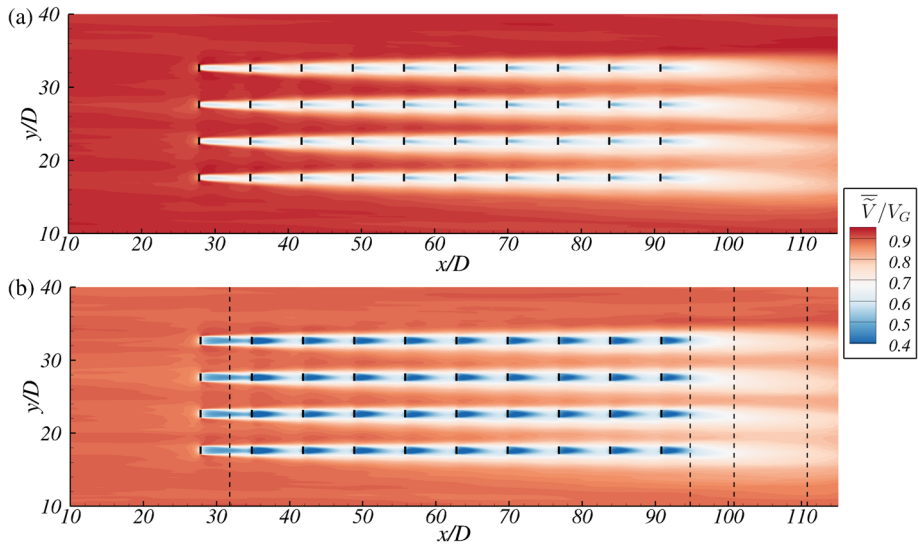


Fig. 14 Horizontal contours of time-averaged horizontal velocity for the CNBL case from the AMD simulation at **a** top-tip height ($z/z_h = 1.5$) and **b** hub height ($z = z_h$). Dashed lines indicate the locations where the spanwise distributions of velocity are extracted

After analyzing the vertical profiles of the ABL flow, we study the turbine wakes through the contours of horizontal wind speed obtained at top-tip height and hub height (Fig. 14). The time-averaged flow field developed over the wind-farm region shows negligible wake turning due to wind veer and Coriolis forcing at both top-tip height and hub height compared to the SBL case (Fig. 6). Further details of the wakes are provided in Fig. 15, where all simulations provide similar normalized time-averaged streamwise velocity profiles downstream of the first row (Fig. 15a). Further downstream of the wind farm (Fig. 15b–d), the velocity profiles within the wake region exhibit limited variability among the SGS models, with the LASDD model showing a higher wake recovery rate. This accelerated recovery is attributed to a higher inflow turbulence intensity of 7.50% for LASDD compared to 5.53% for the other models (Fig. 12d).

The streamwise development of the flow field, along and downstream of the wind farm, is presented in Fig. 16. The streamwise distribution of the normalized jet speed remains nearly constant across the wind farm, with a marginal increase of 0.6% toward the farm's end (Fig. 16a), accompanied by slightly higher turbulence intensity at jet height (Fig. 16b). The jet speed remains unchanged due to the high altitude of the supergeostrophic jet relative to the farm canopy. At hub height, all simulations show a similar distribution of the normalized speed (Fig. 16c), with the exception of the LASDD model, which exhibits a higher wake recovery attributed to its higher turbulence intensity (Fig. 16d). All simulations predict a consistent wind speed evolution over the wind farm, albeit with minor variations in the inflow values (Fig. 16e). The higher inflow turbulence intensity at hub height estimated by the LASDD model (Fig. 16d) results in lower levels of turbine-induced turbulence (Fig. 16f). Near the bottom wall, at $z = z_1$, there is a nearly 5% deceleration of wind speed after the first and second rows, with the WALE model predicting the lowest recovery rate after the fifth row (Fig. 16g). All simulations exhibit a nearly 15% increase in turbulence intensity at $z = z_1$ (Fig. 16h) due to turbine operation. The vertical turbulent transport of mean kinetic

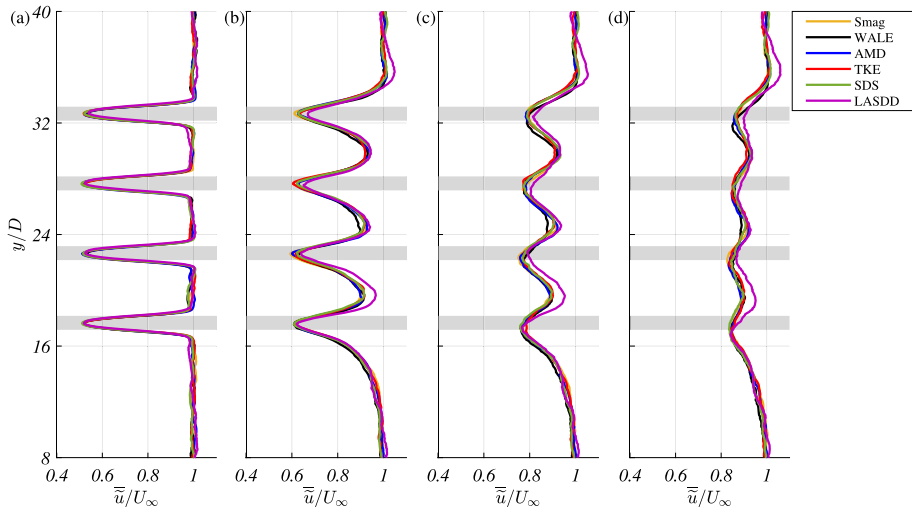


Fig. 15 Spanwise distribution of the time-averaged streamwise velocity $\tilde{u}$ at hub height for the CNBL wind-farm flow at a distance: **a** $4D$ downstream of the first row of turbines, **b** $4D$, **c** $10D$, and **d** $20D$ downstream of the last row of turbines (see dashed lines in Fig. 14b). The profiles are normalized by horizontal and time-averaged wind speed obtained from the precursor simulation $\left(U_{\infty} = \left\langle \tilde{V}(z_h) \right\rangle_{pre} \right)$

energy in the region beneath the turbine rotors is lower in the CNBL case (not shown here for brevity) compared to the SBL, due to the limited ability of the turbines to extract momentum from the higher supergeostrophic jet.

We finalize our analysis of the CNBL case by evaluating the impact of the SGS models on the farm performance. The total farm power production P_F is used as an overall estimate of the farm performance, the AMD and SDS models predict a P_F value of 28.5 MW, while the Smag., WALE, TKE, and LASDD models predict power production values that are 2.46%, 5.96%, 1.40%, and 7.02% lower, respectively. Further details of the flow physics governing power production and wake recovery are provided by an energy budget analysis over the different rows. The first-row power production $|P|_{row=1}$ is used to normalize the various terms in Eq. (54). The AMD, TKE, and SDS models predict a $|P|_{row=1}$ value of 5.84 MW, while the Smag., WALE, and LASDD models yield reduced estimates of 5.71 MW, 5.57 MW, and 5.28 MW, respectively (Table 7). Additionally, the AMD, TKE, and SDS models predict the kinetic energy flux across the west boundary of the first row of turbines $|E|_{k,W row=1}$ (Appendix D) to be approximately 58.03 MW. In comparison, the Smag., WALE, and LASDD models predict lower values of 56.8 MW, 54.7 MW, and 52.4 MW, respectively. For the first row of turbines, the discrepancies in $|E|_{k,W row=1}$ and $|P|_{row=1}$ are primarily due to variations in inflow wind speeds (Figs. 12a and 16e). Furthermore, the turbulent transport of kinetic energy over the inflow west surface, $|T|_{t,W row=1}$, is predicted to be around 0.358 MW by most models, with the exceptions of WALE and LASDD, where the latter particularly increases this contribution to 0.591 MW. After discussing the performance of the first row, we proceed with the budget across the different farm rows. For the CNBL case, the budget of kinetic energy across the wind-farm rows exhibits limited sensitivity to the SGS model. In terms of normalized power production (Fig. 17a), all simulations predict a nearly 55% reduction in power for the secondary rows. The kinetic energy flux remains nearly constant between the second and seventh rows before beginning to decay (Fig. 17b), which contrasts

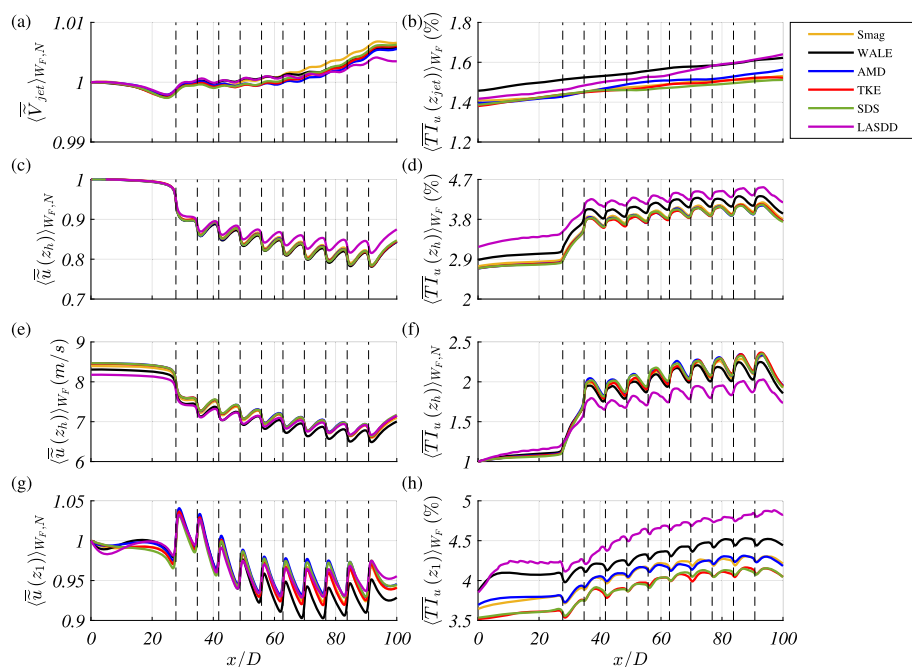


Fig. 16 Streamwise distribution from the CNBL case for the spanwise (over the $20D$ region surrounding the farm) and time-averaged **a** normalized jet speed, **b** streamwise turbulence intensity at jet height, **c** and **e** normalized and absolute streamwise velocity at hub height, **d** and **f** absolute and normalized streamwise turbulence intensity at hub height, and **g** and **h** normalized streamwise velocity and streamwise turbulence intensity at the first grid cell off the wall

Table 7 Results of farm power production P_F ; first-row power production $|P|_{row=1}$; kinetic energy flux $|E|_{k, W row=1}$; and turbulent transport $|T|_{t, W row=1}$ across the west surface of the first row for the CNBL wind-farm flow

	Smag.	WALE	AMD	TKE	SDS	LASDD
P_F (MW)	27.8	26.8	28.5	28.1	28.5	26.5
$ P _{row=1}$ (MW)	5.71	5.57	5.84	5.85	5.82	5.28
$ E _{k, W row=1}$ (MW)	56.8	54.7	58.1	58.0	58.0	52.4
$ T _{t, W row=1}$ (MW)	0.358	0.410	0.360	0.359	0.354	0.591

with the SBL case, where an increase in this term's contribution to the budget is observed after the third row, peaking around the fourth or fifth row depending on the SGS model (Fig. 10). The normalized turbulent transport, flow work, geostrophic forcing, and dissipation are lower in the CNBL case compared to the SBL case, although the trends in the profiles are similar to those predicted in the SBL.

5 Discussion

Large-eddy simulations of the SBL precursor flow are conducted using three different grids: D6, D12, and D45, providing 6, 12, and 45 grid cells across the turbine diameter, respectively.

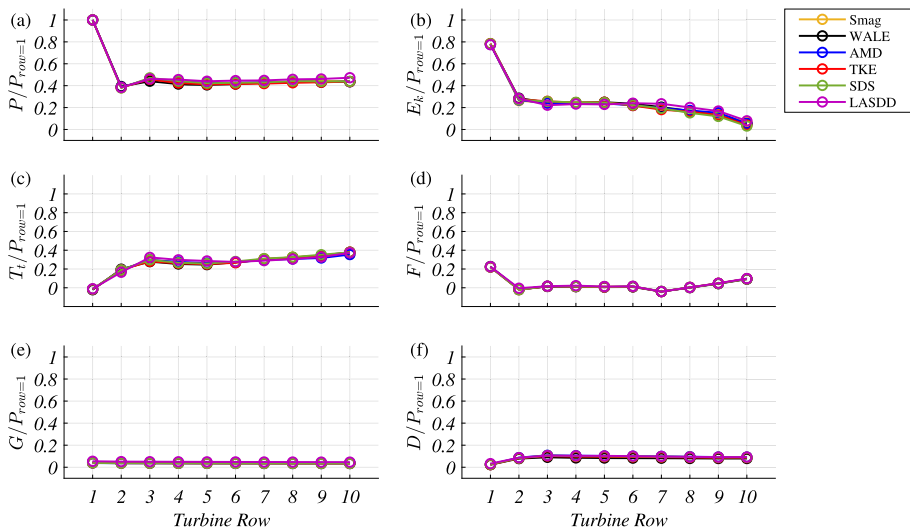


Fig. 17 Row variation of the kinetic energy budget terms (Eq. 54): **a** turbine power production P ; **b** kinetic energy flux E_k ; **c** turbulent transport T_t ; **d** flow work F ; **e** geostrophic forcing G ; and **f** dissipation D , normalized by the power production of the first row $P_{row=1}$, using the different SGS models for the CNBL case

Generally, the results indicate a pronounced sensitivity to both grid spacing and SGS model, particularly for the D6 and D12 grids (Beare et al. 2006; Sullivan et al. 2016; Dai et al. 2020; Wurps et al. 2020). This sensitivity is attributed to the low turbulent mixing characteristic of stable conditions, which results in relatively small turbulence length scales compared to other stability conditions (Doubrawa et al. 2020; Stoll et al. 2020; Hodgson et al. 2023; Syed et al. 2023). In the subsequent discussion, the performance of the six SGS models is evaluated, highlighting the key findings related to each model.

The Smagorinsky model exhibits an over-dissipative behaviour in coarse grids due to the adoption of a constant model coefficient and a fixed SGS Prandtl number. The inversion layer is not captured on the D6 and D12 grids; however, even with only 77% of the turbulent kinetic energy resolved by the D12 grid, key variables such as jet height and friction velocity are predicted with deviations from the D45 grid results of 2% and 8%, respectively. On the finest grid, the inversion layer, jet speed, and height are all consistent with higher-order pseudo-spectral LES and the other evaluated SGS models.

The WALE model demonstrates relatively low dissipation (Fig. 4) (Sharafabadi and Fathali 2023), with a higher proportion of turbulent kinetic energy being resolved by the grid (i.e., 93%, independent of grid size), which is associated with increased momentum and buoyancy fluxes (Figs. 4 and 18). When using the D45 grid, the WALE model's results closely align with those of other SGS models. However, it shows high sensitivity to grid size, with jet height and friction velocity being 19% and 13% greater, respectively, on the D12 grid compared to the D45 grid results. Moreover, models that assume a constant SGS Prandtl number (such as Smagorinsky and WALE) exhibit poor performance in capturing the temperature profile in the SBL case unless a fine grid is used (Fig. 3).

The AMD, TKE, and SDS models all provide similar predictions for the SBL profiles, aligning closely with those from a higher-order pseudo-spectral LES solver and showing low sensitivity to grid size, particularly the AMD and SDS models (Abkar and Moin 2017). When

using the coarsest D6 grid, despite the SDS simulation resolving only 49% of the turbulent kinetic energy, it marginally outperforms the TKE model, which introduces less dissipation near the ground with higher shear stress and heat flux (Fig. 4m). The D12 grid is sufficiently fine for these three SGS models to accurately capture the profile of all flow variables, with jet height and friction velocity differing by less than 5% from the finest D45 grid data.

The LASDD model introduces the least SGS dissipation compared to the other models (Fig. 4) (Beare et al. 2006; Sharafabadi and Fathali 2023). Consequently, simulations using the LASDD model exhibit higher ratios of resolved to total turbulent kinetic energy (i.e., 99% on all grid sizes) and predict highest values of momentum and buoyancy fluxes (Figs. 4 and 18). The lack of added eddy viscosity by the LASDD results in artificially elevated turbulence intensity levels, causing shear stresses to be largest on the coarsest grid and to diminish progressively as the grid is refined. The other models exhibit opposite behavior, as reducing grid size results in a larger amount of resolved turbulence. Even on the finest D45 grid, there are disparities between the LASDD and other models, with a larger jet height and higher turbulence values predicted by the LASDD model. The disparity in results is more pronounced on coarser grids, with a 22% and 12% over-prediction of jet height and friction velocity, respectively, on the D12 grid compared to the D45 grid. This indicates the high sensitivity of LASDD performance to grid size. Notably, Xie et al. (2015) have shown that the performance of the dynamic Smagorinsky models within a finite-difference framework is highly sensitive to grid size, convection scheme, and type of test filter. In DOFAS, box filters of widths 2Δ and 4Δ are used, whereas sharp spectral filters are used in pseudo-spectral codes (Germano et al. 1991). This sensitivity may account for the discrepancies observed between the results of the present study and those of another study that applied the LASDD model in pseudo-spectral solvers (Gadde and Stevens 2021b).

Compared to predictions for the SBL, predictions for the CNBL are notably consistent across all SGS models, aligning closely with one another and with a higher-order pseudo-spectral LES solver (Hodgson et al. 2023). However, the WALE and LASDD models predict slightly larger, hub-height turbulence intensities (6.18–7.50% compared to 5.53% predicted by the other four SGS models), and greater wind veer over the rotor height and geostrophic wind direction values (Table 6). Similar to the SBL case, these models also yield the smallest ratios of SGS dissipation to shear production (Fig. 12g). Notably, the Smag. model with the D12 grid successfully predicts the potential temperature profile (Fig. 12c), in contrast to its performance in the SBL case, where it yields poorer agreement (Fig. 3h). The consistency of the CNBL predictions across different SGS models can be attributed to the larger turbulent length scales present in the CNBL compared to those in the SBL (Peña 2019; Doubrawa et al. 2020; Stoll et al. 2020; Hodgson et al. 2023; Syed et al. 2023). Sullivan et al. (2016) recommend that the grid size should satisfy $z_i/\Delta z > 60$ to accurately resolve boundary-layer physics. This criterion is met in the CNBL scenario with 12 grid cells across the turbine diameter, yielding $z_i/\Delta z = 72$. In contrast, for the SBL case, which features a substantially lower boundary-layer height, the D12 grid corresponds to $z_i/\Delta z = 21$. Therefore, for the stable case, the condition of $z_i/\Delta z > 60$ is only satisfied by the D45 grid, resulting in $z_i/\Delta z = 81$. Accordingly, for wind-farm simulations in which very fine grid sizes are not required for wind-turbine modeling, it is advantageous to employ SGS models that exhibit low sensitivity to grid size (e.g., AMD) and preferably achieve grid independence while minimizing computational cost, thereby reducing uncertainty. In this study, a resolution of 12 grid cells across the turbine diameter is deemed sufficient for these models to accurately predict the stable boundary layer, achieving precision comparable to that obtained with finer grids and higher-order LES solvers. Consequently, the AMD and SDS models are identified as the most

suitable candidates for simulating wind-farm flows under stable conditions characterized by shallow boundary layers, and thus were used with the D12 grid for the wind-farm modelling.

For the wind farm operating under stable conditions, the vertical profiles of velocity deficit, which reflect the wind farm's wake signature within the affected region, are similar for the Smag., AMD, and SDS models, showing a maximum deficit of 19%. In contrast, the WALE and LASDD models predict a maximum deficit of 15.5%, relating to their lower estimates of farm power. The Smag. and AMD models exhibit the largest changes in wind direction over the vertical extent of the turbine rotors, accompanied by the highest increase in resolved shear stress, with a maximum at top-tip height (Fig. 5). Conversely, the WALE and LASDD models predict the lowest farm-induced shear stresses, along with the smallest changes in wind direction and potential temperature over the vertical extent of the turbine rotors.

Another key aspect in analyzing wind-farm flows is the per-row wake recovery. All SGS models predict similar profiles of normalized velocity downstream of the first row of turbines (Fig. 7), despite variations in inflow wind speed and power output from this initial row. Further downstream, the wakes predicted by the Smag., AMD, TKE, and SDS models exhibit similar patterns, whereas the WALE and LASDD models show higher recovery rates due to their higher inflow turbulence (Fig. 8c, d). Additionally, these models tend to predict higher power production in the downstream rows compared to the first row, contrasting with the predictions made by the other SGS models (Fig. 10a). Overall, the AMD, Smag., and SDS models forecast a power output of 25 MW, whereas the WALE, TKE, and LASDD models predict substantial reductions in performance of 26.3%, 12.7%, and 33.9%, respectively.

The different SGS models vary in their predictions of the interaction between the supergeostrophic jet and the wind farm for the SBL case. The Smag., AMD, and SDS models predict a 10% reduction in jet speed, whereas the WALE and LASDD models predict decreases of 4.75% and 2.75%, respectively. This disparity is attributable to the WALE and LASDD models' tendency to overestimate jet height relative to the other models, which limits the vertical turbulent transport of mean kinetic energy from the jet to both the farm and the region below it (Fig. 9). Moreover, the limited vertical turbulent transport of mean kinetic energy results in limited flow acceleration beneath the turbine rotors (Figs. 5a and 8g). Consequently, the WALE and LASDD models, which predict higher supergeostrophic jets, also predict negligible flow acceleration beneath the turbine rotors compared to other models, which predict a flow acceleration of 10–15% near the bottom wall (Fig. 5a). Results from the wind-farm flow simulations under CNBL conditions reveal a consistent performance across all SGS models given that atmospheric turbulence and length scales are larger. The kinetic energy budget exhibits no substantial differences among the models, and the variations in power predictions remain within $\pm 5\%$. However, differences emerge in the streamwise evolution of turbulence intensity, with the LASDD model predicting elevated turbulence intensity at hub height and reduced turbulence intensity near the ground.

A number of recommendations can be drawn from the present results, applicable to numerical codes employing finite-difference, finite-volume, or finite-element discretization methods.

- SGS models that assume a constant SGS Prandtl number, such as the Smag. and WALE models, show limited performance in capturing the temperature profile in SBL conditions unless a very fine grid is used.
- The AMD, TKE, and SDS models consistently deliver favourable results across both thermal stability conditions and various grid sizes within the SBL. Among these models, the AMD and SDS models exhibit a more consistent prediction of mean vertical profiles over the different grid sizes.

- There are no substantial differences between the TKE and SDS models; however, the SDS model is generally preferred in CNBL and moderately stable cases because it assumes a balance between production, dissipation, and buoyancy terms, thereby eliminating the need to solve an additional equation for SGS kinetic energy, as required by the TKE model. However, the SDS model faces challenges in highly stable cases, as it provides zero eddy viscosity when the Richardson number exceeds a certain cutoff value (Allaerts 2016).
- The LASDD model is computationally intensive (Appendix C) and highly dependent on grid size. It exhibits notable discrepancies compared to other models under both SBL and CNBL conditions even for a well-resolved fine grid, with high turbulence intensity associated with its low eddy viscosity.
- The ratio of resolved-to-total turbulent kinetic energy is not a reliable indicator of the SGS model performance (Wurps et al. 2020), as the LASDD and WALE simulations resolve 99% and 93% of the turbulent kinetic energy across all grid sizes, yet results from coarse grids substantially deviate from the finest grid data. Conversely, the SDS model with the coarse D6 grid resolves 49% of the turbulent kinetic energy while providing accurate predictions of shear stresses and heat fluxes.

6 Conclusions

Any explicit large-eddy simulation (LES) solver requires a subgrid-scale (SGS) model to account for the unresolved stresses and fluxes at scales smaller than the filter width, typically taken as the grid size. Various SGS models have been developed for atmospheric boundary-layer flows. However, the suitability of these models for use in LES of wind-farm flows operating in thermally stratified atmospheric conditions has not been considered in published literature. The present study evaluates the performance of six widely used SGS models in the LES of a wind-farm flow considering a stable boundary layer (SBL) and a conventionally-neutral boundary layer (CNBL). This evaluation is conducted on a base-line grid that provides 12 elements across the turbine diameter (i.e., the D12 grid), thereby meeting the resolution requirements of the actuator-disc method, which is widely adopted in wind-farm simulations. The simulations are conducted using the finite-difference code DOFAS, comparing the standard Smagorinsky (Smag.), wall-adapting local eddy-viscosity (WALE), anisotropic minimum-dissipation (AMD), one-equation turbulent kinetic energy (TKE), stability-dependent Smagorinsky (SDS), and Lagrangian-averaged scale-dependent dynamic (LASDD) models. Validation of DOFAS results with those from pseudo-spectral LES codes are also presented.

For the SBL precursor simulations, the D12 grid results are assessed relative to finer D45 and coarser D6 grids, with 45 and 6 grid cells across the turbine diameter, respectively. The SBL precursor flow results show a varying performance of each SGS model on the grid sizes employed. For the D45 grid, providing over 80 cells within the boundary-layer height, the profiles of all flow variables are predicted with similar accuracy across all SGS models, indicating that most of the governing flow scales are resolved. However, the LASDD simulation predicts a slightly elevated supergeostrophic jet, accompanied by larger resolved turbulence. At coarse grid sizes (i.e., D6 and D12 grids), Smag. fails to capture the inversion layer due to its highly dissipative nature. Simulations using the LASDD model, and to a lesser extent, the WALE model, exhibit greater disparity compared to the other SGS model simulations, predicting a marginally higher inversion layer and supergeostrophic jet, as well

as overpredicting values of surface heat flux, friction velocity, turbulent kinetic energy, and Obukhov length. These discrepancies persist despite these two models resolving over 99% of the turbulent kinetic energy, whereas the AMD model achieves greater accuracy compared to results from a pseudo-spectral LES code with only 80% of the turbulent kinetic energy resolved by the grid. The AMD, TKE, and SDS models, using the D12 grid, accurately predict the averaged vertical profiles of wind speed, wind direction, potential temperature, shear stresses, turbulence intensity, and heat fluxes within the SBL, with deviations of only 1–5% from the finest grid results. The AMD and SDS models exhibit minimal sensitivity to grid size. In the CNBL flow, the D12 grid is sufficient to resolve the boundary layer, with results showing minimal sensitivity to the SGS model used. Only minor deviations are noted between the models in the CNBL case, with the LASDD model slightly underestimating jet height and overestimating turbulence intensity.

The performance of the 40-turbine wind farm is analyzed using the D12 grid, as required by the actuator-disc method. In the SBL conditions, the AMD and SDS models produce nearly identical power output of 25 MW. The WALE and LASDD models predict a higher supergeostrophic jet and reduced vertical turbulent transport of mean kinetic energy from the jet to the farm, along with lower inflow wind speeds, resulting in about 30% less power production compared to other models. In the CNBL conditions, predictions of farm performance exhibit less sensitivity to the choice of SGS model. The AMD, TKE, and SDS models predict similar power output of 28.5 MW, whereas the Smag., WALE, and LASDD models predict slightly lower power output.

The present analysis demonstrates that LES conducted using grids with sizes that are adequate for the actuator-disc method, while also providing more than 60 grid cells through the boundary-layer height, can capture the characteristics of thermally stratified flows, irrespective of the SGS model used. In stable conditions characterized by low atmospheric turbulence and a shallow boundary layer, providing 80 grid cells across the boundary-layer height can be computationally impractical. Consequently, the SGS model must be chosen carefully in these scenarios, as it influences LES results by affecting both the precursor simulations and the turbulence within the farm that drives wake recovery. Among the six SGS models evaluated, the AMD and SDS models exhibit low sensitivity to grid size. Furthermore, they predict velocity, turbulence, and heat flux profiles with accuracy comparable to higher-order schemes. Moreover, the AMD model proves particularly advantageous due to its ease of implementation, allowing for variation in SGS eddy viscosity and eddy diffusivity without the need for parameter tuning. This characteristic underscores the high potential for the AMD model to be used in the LES of wind-farm flows under stable and conventionally-neutral conditions.

A Computation of Body Forces in DOFAS

A.1 Mean Wind-Angle Controller

A mean wind-angle controller is used to rotate the entire flow field in the precursor simulation (Sescu and Meneveau 2014; Allaerts 2016) such that the mean flow is aligned with the wind farm at hub height. The forcing term f_i^{wc} is given by:

$$f_i^{wc} = \varepsilon_{ij3} \tilde{u}_j \omega^{wc}, \quad (19)$$

with ω^{wc} denoting the angular speed given by:

$$\omega^{wc} = \hat{\omega} + \beta^{wc} (\phi^{wc,n} - \phi^{des}), \quad (20)$$

where β^{wc} is taken to be $1/7200 \text{ s}^{-1}$ (Allaerts 2016); ϕ^{des} is the desired mean flow angle at a certain height z^{wc} (taken as the turbine hub height z_h for the present study); $\phi^{wc,n}$ is the actual mean flow angle at this height at the current time step given by:

$$\tan(\phi^{wc,n}) = \frac{\langle \tilde{v}(z^{wc}) \rangle}{\langle \tilde{u}(z^{wc}) \rangle}; \quad (21)$$

$\hat{\omega}$ is the time-filtered angular-speed, where $\hat{\cdot}$ denotes time filtering to average rapid turbulent fluctuations using a first-order filter (Allaerts 2016) as follows:

$$\hat{\omega} = \left(1 - \frac{\Delta t}{\tau_f + \Delta t}\right) \hat{\omega}^{t-\Delta t} + \left(\frac{\Delta t}{\tau_f + \Delta t}\right) \omega; \quad (22)$$

the time increment is denoted by Δt ; $\tau_f = 200 \text{ s}$ is the filter constant (Allaerts 2016); ω is the instantaneous angular speed at any time step given by:

$$\omega = \frac{\phi^{wc,n} - \phi^{wc,n-1}}{\Delta t}; \quad (23)$$

and $\hat{\omega}^{t-\Delta t}$ is the time-filtered angular speed from the previous time step. The mean wind controller is turned off after the precursor simulation has reached the quasi-steady state.

A.2 Fringe-Region Forces

To simulate a finite wind farm with turbulent ABL inflow conditions, a precursor–concurrent technique is adopted (Stevens et al. 2014). A precursor (boundary-layer flow) simulation with no turbines (Fig. 1a) is run first. This precursor simulation is driven a mean pressure gradient that is related to the geostrophic wind speed (Eq. (4)), with periodic boundary conditions applied in both horizontal directions. The simulation runs until the flow reaches a quasi-steady state [i.e., the root mean square of the time-averaged wind speed over two consecutive hours is less than 0.1 m/s (Beare et al. 2006)] denoted by t_{SS} . Over this period (i.e., $0 \leq t \leq t_{SS}$), a mean wind-angle controller is used to rotate the entire flow field such that the mean flow is aligned with the wind farm at hub height (Sescu and Meneveau 2014; Allaerts 2016). The flow variables from the precursor simulation during the interval $t_{SS} \leq t \leq t_F$, with t_F being the final simulation time, are used as inflow conditions for the concurrent wind-farm simulation. The concurrent simulation is initialized with the instantaneous flow field from the precursor simulation obtained at t_{SS} . The total concurrent simulation time is $t_F - t_{SS}$. A fringe region (Fig. 1b) is applied in the concurrent simulation to drive it using the precursor flow field as an inflow. The fringe forces f_i^{frx} and f_i^{fry} are volume forces added in the fringe regions (Lanzilao and Meyers 2022), to impose the inflow velocities denoted by $U_{in,i}$ and $V_{in,i}$ for the x and y directions, respectively. f_i^{frx} and f_i^{fry} are given by:

$$f_i^{frx} = -h_x(x) (\tilde{u}_i - U_{in,i}); \quad (24)$$

$$f_i^{fry} = -h_y(y) (\tilde{u}_i - V_{in,i}), \quad (25)$$

where each of $h_x(x)$ and $h_y(y)$ is a one-dimensional non-negative function that is non-zero in the fringe region only. $h_x(x)$ and $h_y(y)$ are given by:

$$h_x(x) = h_{max} \left[F \left(\frac{x - x_s^{fr}}{\delta_s^h} \right) - F \left(\frac{x - x_e^{fr}}{\delta_e^h} + 1 \right) \right]; \quad (26)$$

$$h_y(y) = h_{max} \left[F \left(\frac{y - y_s^{fr}}{\delta_s^h} \right) - F \left(\frac{y - y_e^{fr}}{\delta_e^h} + 1 \right) \right], \quad (27)$$

where the function F is provided as:

$$F(\xi) = \begin{cases} 0, & \xi \leq 0 \\ \frac{1}{1 + \exp\left(\frac{1}{\xi - 1} + \frac{1}{\xi}\right)}, & 0 < \xi < 1 \\ 1, & \xi \geq 1 \end{cases}. \quad (28)$$

The parameters x_s^{fr} and x_e^{fr} denote the start and end of the fringe region in the x direction; and y_s^{fr} and y_e^{fr} are the start and end of the fringe region in the y direction, respectively. The parameters h_{max} ; δ_s^h ; and δ_e^h have been chosen according to Lanzilao and Meyers (2022).

A.3 Rayleigh-Damping Forces

Irregular topography and spatial variations in forcing, such as those from wind farms, can induce atmospheric gravity waves by causing vertical displacement of the flow. These gravity waves propagate upward and are advected downstream by the flow (Allaerts 2016). To prevent these waves from reflecting back into the region of interest and causing numerical errors or instabilities in computational simulations, a Rayleigh-damping layer is applied at the upper boundary of the computational domain (Klemp and Lilly 1978; Klemp and Durran 1983; Lanzilao and Meyers 2022). This layer effectively absorbs the gravity waves, ensuring the accuracy and stability of the simulation. The Rayleigh-damping force f_i^{rd} is given by:

$$f_i^{rd} = -v^{rd}(z) (\tilde{u}_i - U_{G_i}), \quad (29)$$

where (U_{G_1}, U_{G_2}) are the horizontal components of the geostrophic wind speed; U_{G_3} is zero; and $v^{rd}(z)$ is the Rayleigh function given by:

$$v^{rd} = \begin{cases} 0, & z < L_z - L_z^{rd} \\ \check{v} \left[1 - \cos \left(\frac{\pi}{s} \frac{z - (L_z - L_z^{rd})}{L_z^{rd}} \right) \right], & z \geq L_z - L_z^{rd} \end{cases}, \quad (30)$$

in which $\check{v}$ is a damping coefficient to control the force magnitude; s controls the function gradient along the vertical direction; L_z is the vertical length of the domain; and L_z^{rd} is the length of the Rayleigh-damping layer.

A.4 Turbine Forces

The forces from the wind turbine blades f_i^{turb} can be resolved in DOFAS either with the actuator-line method (Yang et al. 2014; Ouro et al. 2019; Willden et al. 2023) or, as in the present study with an actuator-disc method. The actuator-disc method has been used in similar wind-farm simulation studies proving accurate to predict the far-wake structure behind turbines (Calaf et al. 2010; Gadde and Stevens 2021a; Stieren and Stevens 2022; Sharafabadi and Fathali 2023). In the non-rotating ADM (Calaf et al. 2010), the flow is assumed normal to the disc and the tangential forces are neglected, with the total thrust force of a single turbine calculated as:

$$F^{turb} = -\frac{1}{2} \rho C'_T \langle \hat{u} \rangle_d^2 \left(\frac{\pi}{4} D^2 \right), \quad (31)$$

where D is the turbine rotor diameter; $\langle \hat{u} \rangle_d$ is the disc-averaged and time-filtered velocity normal to the rotor plane given by:

$$\langle \hat{u} \rangle_d = \left(1 - \frac{\Delta t}{\tau_f + \Delta t} \right) \langle \hat{u} \rangle_d^{t-\Delta t} + \left(\frac{\Delta t}{\tau_f + \Delta t} \right) \langle \tilde{u} \rangle_d, \quad (32)$$

where $\langle \cdot \rangle_d$ denotes spatial averaging over the disc; the filter constant τ_f is taken to be 200 s; $\langle \tilde{u} \rangle_d$ is the instantaneous disc-averaged velocity at a certain time step; and $\langle \hat{u} \rangle_d^{t-\Delta t}$ is the disc-averaged and time-filtered velocity from the previous time step. The modified thrust coefficient C'_T is given by:

$$C'_T = \frac{C_T}{(1-a)^2}, \quad (33)$$

with C_T is a thrust coefficient and a is the induction factor. Lagrangian points are distributed over the turbine rotor and the force per unit mass assigned to each point is given by:

$$f_L^{turb} = -\frac{1}{2\Delta x} C'_T \langle \hat{u} \rangle_d^2, \quad (34)$$

where the thickness of the actuator disc in the flow direction is taken to be Δx . The turbine forces per unit mass f_L^{turb} are mapped to the neighbouring Eulerian fluid grid points as follows:

$$f_i^{turb}(\vec{x}) = \frac{\hat{n}_i}{\Delta V} \int \int \int f_L^{turb} \delta_h(\vec{x} - \vec{x}') dx' dy' dz', \quad (35)$$

where $(\hat{n}_1, \hat{n}_2, \hat{n}_3) = (\cos \gamma, \sin \gamma, 0)$; γ is the yaw angle; $\Delta V = \Delta x \Delta y \Delta z$ is the volume of the fluid grid cell; δ_h is a 4-point piecewise delta interpolation function (Yang et al. 2009); $\vec{x} = (x, y, z)$ is the coordinates vector of the grid points; and $\vec{x}' = (x', y', z')$ is the coordinates vector of the Lagrangian points distributed over the turbine disc area. Finally, the yaw angle of each turbine is determined by an active local yaw controller, which rotates the turbine rotor to be normal to the flow field $1D$ upstream of the turbine using time-filtered values over a window of 10 min (Allaerts 2016).

B Subgrid-Scale Models in DOFAS

B.1 Smagorinsky Model

The standard Smagorinsky model (Smagorinsky 1963) has been used in many applications for its simplicity and relatively low computational cost. Analogous to Prandtl's mixing-length model used in RANS simulations, the Smagorinsky model assumes that the SGS eddy viscosity ν_{SGS} (units m^2/s) can be expressed using a length scale multiplied by a velocity scale ($\nu_{SGS} = l_{SGS} \cdot v_{SGS}$). In LES, the eddies larger than the filter size are simulated, while the smaller eddies are parameterized using the SGS model. The filter size is usually related to the grid size, therefore, it is reasonable to relate the length scale to the average grid size ($l_{SGS} = C_s \Delta$), where C_s is the model coefficient and $\Delta = \sqrt[3]{\Delta x \Delta y \Delta z}$. The velocity scale is taken to be the length scale multiplied by the characteristic filtered rate of strain $\tilde{S} = (2\tilde{S}_{ij}\tilde{S}_{ij})^{0.5}$. This yields the following expression for the eddy viscosity in the Smagorinsky SGS model:

$$\nu_{SGS} = C_s^2 \Delta^2 \tilde{S}. \quad (36)$$

Equation (36) shows that ν_{SGS} is always greater than zero. This means that the energy is dissipated from the large eddies to the smaller ones only (i.e., this model does not allow backscatter). Finally, the Smagorinsky model has one universal length scale to model all eddies. This represents a drawback when trying to model near-wall eddies, as the size of the eddies reduces compared to the filter (grid) scale. To remedy this, the near-wall damping function proposed by Mason and Thomson (1992) is adopted to reduce the length scale near the wall as follows:

$$l_{SGS}^{-n} = (C_s \Delta)^{-n} + [\kappa (z + z_o)]^{-n}, \quad (37)$$

where n is a damping exponent taken to be 2. Finally, the Smagorinsky model with near-wall damping provides the SGS eddy viscosity as:

$$\nu_{SGS} = l_{SGS}^2 \tilde{S}. \quad (38)$$

To obtain the SGS eddy diffusivity ($\Gamma_{SGS} = \nu_{SGS}/Pr_{SGS}$), a value for the SGS Prandtl number needs to be specified. For the Smagorinsky model, different values of C_s and Pr_{SGS} have been proposed in the literature. The value of C_s ranges from 0.10 to 0.21, while Pr_{SGS} ranges from 0.44 to 0.70. $C_s = 0.17$ and $Pr_{SGS} = 0.5$ have been used by many authors (Gadde et al. 2021a) and have been reported to show acceptable results in related types of flow, hence adopted in the present study with damping exponent $n = 2$.

B.2 WALE Model

The choice of the characteristic filtered rate of strain $\tilde{S}$ to obtain a velocity scale for the Smagorinsky model (Eq. (36)) is arbitrary. The WALE model uses the traceless symmetric part of the square of the velocity gradient tensor to better represent the velocity scale (Nicoud and Ducros 1999), taking into account not only the strains but also the rotations of the smallest-scale resolved eddies. In the WALE model, the SGS eddy viscosity is given by:

$$\nu_{SGS} = C_w^2 \Delta^2 \frac{(\tilde{S}_{ij}^d \tilde{S}_{ij}^d)^{1.5}}{(\tilde{S}_{ij} \tilde{S}_{ij})^{2.5} + (\tilde{S}_{ij}^d \tilde{S}_{ij}^d)^{1.25}}, \quad (39)$$

where $C_w = 0.46$ is the WALE coefficient and $\tilde{S}_{ij}^d$ is the traceless symmetric part of the square of the velocity gradient tensor defined by:

$$\tilde{S}_{ij}^d \equiv \frac{1}{2} \left(\frac{\partial \tilde{u}_k}{\partial x_i} \frac{\partial \tilde{u}_j}{\partial x_k} + \frac{\partial \tilde{u}_k}{\partial x_j} \frac{\partial \tilde{u}_i}{\partial x_k} \right) - \frac{1}{3} \delta_{ij} \frac{\partial \tilde{u}_k}{\partial x_l} \frac{\partial \tilde{u}_l}{\partial x_k}. \quad (40)$$

This model implicitly reduces turbulence near wall boundaries, avoiding the need for near-wall damping functions. The model turns off in a laminar flow (i.e., can capture the transition of the flow from laminar to turbulent). Additionally, it does not require any horizontal-averaging operations (that requires the existence of flow homogeneity). A constant SGS Prandtl number has to be specified to determine the SGS eddy diffusivity, with $Pr_{sgs} = 0.5$ used in the present study.

B.3 AMD Model

Verstappen (2011) has developed the first minimum-dissipation model, based on the concept of obtaining the minimum eddy dissipation required to dissipate the energy of the unresolved

subfilter scales. While Verstappen (2011) has focused on isotropic grids, Rozema et al. (2015) have extended the minimum-dissipation model for anisotropic grids, and Abkar et al. (2016) and Abkar and Moin (2017) have developed an AMD-based model for SGS fluxes. The main requirement of these models is that the energy of the subfilter scales in a filter box must always decrease with time, with the size of the filter box usually deemed as the grid size. The AMD model (Abkar and Moin 2017) provides the following relations for the SGS eddy viscosity and eddy diffusivity, respectively:

$$\nu_{SGS} = \frac{-\left(\hat{\partial}_k \tilde{u}_i\right)\left(\hat{\partial}_k \tilde{u}_j\right) \tilde{S}_{ij} + \delta_{i3}\left(g / \theta_o\right)\left(\hat{\partial}_k \tilde{u}_i\right) \hat{\partial}_k\left(\tilde{\theta}-\langle\tilde{\theta}\rangle\right)}{\left(\partial_l \tilde{u}_m\right)\left(\partial_l \tilde{u}_m\right)} ; \quad (41)$$

and:

$$\Gamma_{SGS} = \frac{-\left(\hat{\partial}_k \tilde{u}_i\right)\left(\hat{\partial}_k \tilde{\theta}\right) \partial_i \tilde{\theta}}{\left(\partial_l \tilde{\theta}\right)\left(\partial_l \tilde{\theta}\right)}, \quad (42)$$

where $\hat{\partial}_i$ is the scaled gradient operator defined as $\hat{\partial}_i \equiv \sqrt{C_i} \Delta x_i \partial_i$; and C_i is a modified Poincare constant. C_i is taken equal to 1/3 if second-order accurate scheme is used in the x_i direction, as in DOFAS, whereas C_i is taken equal to 1/12 if spectral discretization is used. This model is relatively simple and easy to implement, and does not require any averaging processes. No near-wall damping functions are required by the model, and the model turns off in laminar flows. Also, one of the advantages of this model is that it does not require any parameter tuning.

B.4 TKE Model

The TKE model, developed by Deardorff (1980), solves a transport equation for the SGS kinetic energy. The equation takes into consideration the shear production, dissipation, diffusion, and buoyancy effects. After the SGS kinetic energy is evaluated, it is related to the SGS viscosity as follows:

$$\nu_{SGS} = C_k l_{SGS} \sqrt{e_{SGS}}, \quad (43)$$

where C_k is the model coefficient; $e_{SGS} \equiv \frac{1}{2} \tau_{ii}^{SGS}$ is the SGS kinetic energy provided by:

$$\frac{\partial e_{SGS}}{\partial t} + \tilde{u}_j \frac{\partial e_{SGS}}{\partial x_j} = -\tau_{ij}^{SGS} \tilde{S}_{ij} - C_\varepsilon \frac{e_{SGS}^{3/2}}{l_{SGS}} + \frac{g}{\theta_o} q_3^{SGS} + \frac{\partial}{\partial x_i} \left(2\nu_{SGS} \frac{\partial e_{SGS}}{\partial x_i} \right), \quad (44)$$

where C_ε is the dissipation coefficient; and the length scale is reduced in case of stably stratified conditions according to:

$$l_{SGS} = \min \left(\Delta, l_e \right), \quad (45)$$

with l_e denoting a stability-related length scale defined as:

$$l_e = 0.76 \sqrt{\frac{e_{SGS}}{\frac{g}{\theta_o} \frac{\partial \tilde{\theta}}{\partial z}}}. \quad (46)$$

Spectral analysis (Moeng and Wyngaard 1988) has been used to obtain the values of C_k and C_ε as $C_k = 0.1$ and $C_\varepsilon = 0.93$, and these values have been used in the present study. The transport equation (Eq. (44)) is discretized and solved with the rest of the governing equations using the modified Runge–Kutta scheme. For the bottom-wall boundary condition, wall stresses and fluxes are imposed in the production and buoyancy terms, respectively,

according to the Monin–Obukhov similarity theory. Finally, the SGS eddy diffusivity can be obtained as:

$$\Gamma_{SGS} = \left(1 + 2 \frac{l_{SGS}}{\Delta}\right) \nu_{SGS}. \quad (47)$$

It is worth noting that the Smagorinsky model can be derived from the TKE model if a balance is assumed between shear production and dissipation. This balance provides the Smagorinsky coefficient as:

$$C_s = \left(\frac{C_k^3}{C_\varepsilon}\right)^{0.25} \approx 0.18. \quad (48)$$

B.5 SDS Model

In the SDS model (Stevens et al. 2000), the Smagorinsky coefficient is derived from the TKE equation (Eq. (44)) including stability effects, and balance is assumed between the production, dissipation, and buoyancy terms to derive a stability-dependent model. The SGS kinetic energy is given by:

$$e_{SGS} = \frac{C_k}{C_{\varepsilon 1} + \frac{C_{\varepsilon 2} l_{SGS}}{\Delta}} (l_{SGS} \tilde{S})^2 \left(1 - \frac{Ri}{Pr_{SGS}}\right), \quad (49)$$

where Ri is the gradient Richardson number defined as:

$$Ri = \left(\frac{g}{\theta_o} \frac{\partial \tilde{\theta}}{\partial z}\right) / \tilde{S}^2. \quad (50)$$

Substitution from Eq. (49) into Eq. (43) yields:

$$\nu_{SGS} = \left[\left(\frac{C_k^3}{C_{\varepsilon 1} + \frac{C_{\varepsilon 2} l_{SGS}}{\Delta}} \right)^{0.25} l_{SGS} \right]^2 \tilde{S} \sqrt{1 - \frac{Ri}{Pr_{SGS}}}. \quad (51)$$

The SGS eddy diffusivity is calculated as $\Gamma_{SGS} = \nu_{SGS} / Pr_{SGS}$. In this model, the SGS Prandtl number is calculated as:

$$Pr_{SGS} = \frac{1}{1 + 2 \frac{l_{SGS}}{\Delta}}. \quad (52)$$

To avoid the lack of continuity of eddy viscosity, the SGS length scale l_{SGS} is calculated as the geometric mean between the grid size Δ and the stability-dependent length scale l_e (Eq. (46)) (Allaerts 2016), instead of using a minimum function, as the one used in Eq. (45). Accordingly, the length scale is given by:

$$l_{SGS}^{-n} = \Delta^{-n} + l_e^{-n} + [\kappa (z + z_o)]^{-n}. \quad (53)$$

Finally, the model coefficients C_k , $C_{\varepsilon 1}$, $C_{\varepsilon 2}$, C_l , n are given by 0.1, 0.225, 0.705, 0.82, 2, respectively. The SDS model is implemented as follows: (i) l_e is calculated using Eq. (46), with the value of e_{SGS} taken from the previous time increment, (ii) l_{SGS} is calculated using Eq. (53), (iii) ν_{SGS} and Pr_{SGS} are obtained using Eqs. (51) and (52), respectively, and (iv) e_{SGS} is calculated using Eq. (49).

Although the model is relatively simple as it avoids solving the additional transport equation for the SGS turbulent kinetic energy, its major drawback is that when the Richardson number reaches a critical value ($Ri_{cr} = (1 + 2l_{SGS}/\Delta)^{-1}$), then e_{SGS} has a value of zero

and the SGS model is turned off. In other words, this model assumes that in cases of strong thermal stability ($Ri \geq Ri_{cr}$), the energy cascade is turned off and turbulence decays, which has been proven to be physically invalid (Allaerts 2016).

B.6 LASDD Model

The standard Smagorinsky model uses a single constant to account for all SGS eddies in the entire flow field, which stands as its major disadvantage. This constant length scale leads to relatively large SGS stresses existing near solid boundaries or in laminar flows where they should not exist.

Germano et al. (1991) proposed a dynamic SGS model with the model coefficient being calculated locally for each cell at every time step ($C_s = C_s(x_i, t)$), instead of applying a constant value a priori. In the dynamic model, the Germano identity is used to relate the stresses at two different length scales: one is the grid filter width Δ and the other is the test filter width, usually taken to be 2Δ . A sharp spectral test filter is commonly used in pseudo-spectral codes (Germano et al. 1991), whereas a box filter (Balaras and Benocci 1994; Najjar and Tafti 1996) is implemented in DOFAS. The Smagorinsky approximation (Eqs. (11) and (36)) is used to express the stresses at the two different length scales. The relation is averaged [spatially (Lilly 1992) or along pathlines (Meneveau et al. 1996)] to avoid numerical instability, and errors due to the Smagorinsky approximation are minimized to obtain a formula for the dynamic Smagorinsky coefficient.

In the scale-independent version of the dynamic Smagorinsky model (Germano et al. 1991; Meneveau et al. 1996), the model coefficient is assumed to be the same for the two filters (i.e., $C_{s,2\Delta} = C_{s,\Delta}$). This assumption has been removed in scale-dependent models by assuming that the coefficient has a linear profile [i.e., $C_{s,2\Delta}^2 = \alpha C_{s,\Delta}^2$ (Porté-Agel et al. 2000; Porté-Agel 2004; Bou-Zeid et al. 2005)]. To calculate the scale-dependence parameter α , another test filter of width 4Δ is used where $C_{s,4\Delta}^2 = \alpha C_{s,2\Delta}^2$, this provides another equation to calculate α locally. However, this requires filtering at two test filters (i.e., 2Δ and 4Δ) rather than one. A similar approach has been developed to obtain the dynamic SGS diffusivity. Full details about the filtering, clipping, and averaging operations are provided by Stoll and Porté-Agel (2006).

The SGS stresses, obtained using the dynamic Smagorinsky model, vanish in laminar flows and near solid boundaries with the near-wall velocity profile matching the logarithmic profile. However, the LASDD model is more sophisticated than the standard Smagorinsky model as it requires additional averaging and filtering processes for different flow quantities which increases the computational demand. Additionally, the type of the test filter (e.g., box, Gaussian, spectral cutoff filters) substantially impacts the results (Xie et al. 2015), which represents another drawback of the LASDD model.

C Computational Aspects and Grid Sensitivity Analysis

The precursor simulations for the SBL are run with the three different grids (i.e., D6, D12, and D45) on AMD EPYC 7742 cores on UK's ARCHER2 facility. For the D6 grid, 100 cores are used with 1 h of physical time needed on average to run each simulation, except for LASDD, which requires nearly 3 h of physical time. For the D12 grid simulations, 12,800 cores are adopted to run each simulation over nearly 5 h of physical time, with LASDD requiring 16 h of physical time. With the D45 grid, 200 cores are used to run the simulation

Table 8 Average computational time per time step required to compute the SGS model subroutine in DOFAS for the SBL, with the D12 grid setup

	Smag.	WALE	AMD	TKE	SDS	LASDD
Time (s)	0.025	0.0288	0.02618	0.06263	0.05	0.2175

Table 9 Results obtained with the grids D6, D12, and D45, using the different SGS models, of normalized jet speed V_{jet}/V_G ; normalized jet height z_{jet}/z_h ; geostrophic wind direction ϕ_G ; and normalized inversion height z_c/z_h

Model	V_{jet}/V_G			z_{jet}/z_h			$\phi_G (^\circ)$			z_c/z_h		
	D6	D12	D45	D6	D12	D45	D6	D12	D45	D6	D12	D45
Smag.	1.14	1.18	1.20	1.75	1.63	1.79	-17.0	-20.6	-20.8	3.08	2.88	2.30
WALE	1.17	1.18	1.20	2.42	2.13	1.79	-26.2	-25.3	-19.9	2.92	2.46	2.28
AMD	1.17	1.18	1.20	1.75	1.79	1.81	-21.0	-20.2	-20.3	2.08	2.04	2.23
TKE	1.14	1.17	1.19	2.08	1.88	1.81	-22.1	-21.0	-21.0	2.08	2.13	2.19
SDS	1.14	1.17	1.20	1.75	1.71	1.79	-19.2	-19.9	-20.7	1.92	2.04	2.21
LASDD	1.16	1.18	1.20	3.25	2.38	1.94	-28.4	-27.9	-22.4	3.58	3.04	2.10

in 5 h on average, except for LASDD, which needs 16 h of physical time. An estimate for the computational overhead of each SGS model to the LES is provided in Table 8, providing the average computational time required to run each SGS model once for the SBL with the D12 grid setup. Smag. requires marginally the lowest computational time, with WALE and AMD requiring nearly similar computational times. TKE and SDS require nearly 2.5 times and 2 times the Smag.'s computational time, respectively. LASDD requires nearly 9 times the Smag.'s computational time due to the additional filtering and averaging processes. LASDD also has additional storage requirements to obtain 20 Lagrangian-averaged values, which are used to calculate the scale-dependence parameter (Stoll and Porté-Agel 2006). Accordingly, the high computational cost represents a drawback for LASDD.

Table 9 shows horizontal and time-averaged values of the jet speed V_{jet} (i.e., the value of the wind maxima), normalized by the geostrophic wind speed $V_G = (U_{G1}^2 + U_{G2}^2)^{0.5}$; normalized jet height z_{jet}/z_h (where z_{jet} is the height at which the wind speed attains its maximum value); geostrophic wind direction $\phi_G = \langle \tilde{\phi}(L_z) \rangle$, where $\tilde{\phi} = \arctan(\tilde{v}/\tilde{u})$; and normalized inversion height z_c/z_h [i.e., the height at which the mean temperature profile has the maximum gradient (Gadde and Stevens 2021a)], for the three grids using the SGS models.

For the turbulence quantities, Table 10 shows horizontal and time-averaged values of the hub-height streamwise turbulence intensity $\langle \overline{TI}_u(z_h) \rangle = \left(\overline{u'u'}(z_h) \right)^{0.5} / U_\infty$, where $U_\infty = \langle \tilde{V}(z_h) \rangle$ is the horizontal and time-averaged hub-height wind speed; normalized boundary-layer height z_i/z_h ; and normalized friction velocity $u_*/V_G = \sqrt{\langle \overline{\tau}_w \rangle} / V_G$, obtained by the three grids, using the different SGS models.

The analysis of heat fluxes reveals the variations among SGS models, with the Smag., AMD, and SDS models predicting a surface heat flux q_w of approximately -0.011 K m/s using the D6 grid (Fig. 18a and Table 11), whereas the TKE model predicts a slightly higher

Table 10 Results obtained with the grids D6, D12, and D45, using the different SGS models, of the hub-height streamwise turbulence intensity $\langle \overline{T} \overline{I}_u(z_h) \rangle$; normalized boundary-layer height z_i/z_h ; and normalized friction velocity u_*/V_G

Model	$\langle \overline{T} \overline{I}_u(z_h) \rangle$ (%)			z_i/z_h			u_*/V_G		
	D6	D12	D45	D6	D12	D45	D6	D12	D45
Smag.	0.00	3.97	3.67	2.00	1.79	1.75	0.0294	0.0300	0.0319
WALE	7.08	5.84	3.51	2.43	2.19	1.73	0.0373	0.0356	0.0315
AMD	4.06	4.10	3.75	1.91	1.66	1.76	0.0318	0.0305	0.0314
TKE	5.54	4.53	3.83	2.15	1.97	1.77	0.0337	0.0317	0.0321
SDS	3.28	4.33	3.66	1.78	1.81	1.72	0.0310	0.0308	0.0318
LASDD	12.2	6.98	4.88	3.24	2.41	2.01	0.0444	0.0380	0.0338

value of -0.0126 K m/s. The WALE and LASDD models predict larger values compared to the other models, with $q_w = -0.0154$ K m/s and -0.0212 K m/s, respectively. Compared with LES data reported by Gadde et al. (2021a), the WALE and LASDD models predict higher total vertical heat flux values throughout the entire boundary-layer height, and the Smag. model shows a substantial increase at $z/z_h > 1.3$. In the D12 grid results (Fig. 18b), the Smag., AMD, TKE, and SDS models exhibit similar vertical heat flux profiles, predicting $q_w = -0.0107$ K m/s, closely matching the data from Gadde et al. (2021a). However, higher values are seen from the Smag. model for $z/z_h > 1.75$. The WALE and LASDD models still exhibit the largest vertical heat flux over the boundary-layer height, with $q_w = -0.0134$ K m/s and -0.0157 K m/s, respectively. For the D45 grid (Fig. 18c), all models show similar vertical heat flux profiles with $q_w = -0.0104$ K m/s, except for the LASDD model, which predicts higher values with $q_w = -0.0113$ K m/s, representing an 8.8% increase compared to the other simulations. Although the vertical distribution of total heat flux is similar across the Smag., AMD, TKE, and SDS models (Fig. 18a), substantial differences exist between their resolved and SGS components for the D6 (Fig. 18d, g) and D12 grids (Fig. 18e, h). The results obtained with the D45 grid (Fig. 18f, i) show a negligible contribution from the SGS heat flux, and consequently, minimal sensitivity to the choice of SGS model, with the exception of the LASDD model, which predicts higher resolved heat fluxes. Notably, the trend exhibited by the different simulations in predicting the resolved and SGS heat fluxes is consistent with their predictions in predicting the resolved and SGS momentum fluxes (Fig. 4). Finally, Table 11 also presents values of the normalized Obukhov length L_O/z_h for the three grids using different SGS models, showing that the WALE and LASDD models with the coarser grids predict larger Obukhov lengths compared to the D45 grid values, with closer values attained by the AMD, TKE, and SDS models for the D6 and D12 grid sizes.

Figure 19 shows the convergence of (a) jet height z_{jet} and (b) friction velocity u_* , expressed as the percentage deviation from the values obtained using the D45 grid for the SGS models $[(\cdot) - (\cdot)_{D45}] / (\cdot)_{D45}$. Generally, the AMD and SDS models exhibit the least sensitivity to grid size variations. For the D12 grid, the Smag., AMD, TKE, and SDS simulations predict z_{jet} and u_* with deviations of nearly 5% from the values obtained using the finest D45 grid. In contrast, the WALE and LASDD models predict these values with deviations of 15–20% from the D45 grid values, indicating higher sensitivity to grid size.

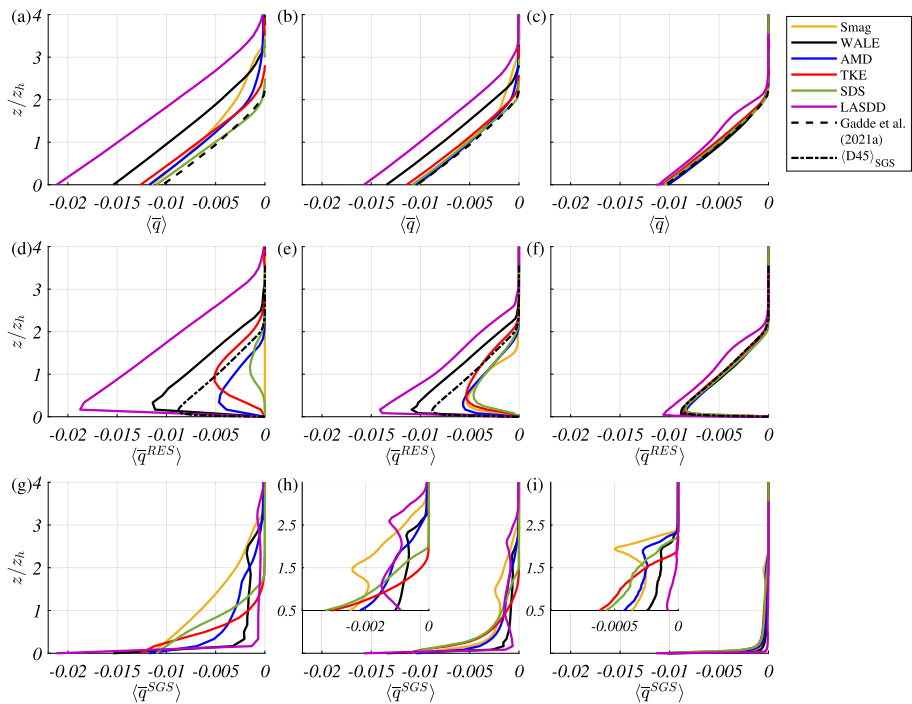


Fig. 18 Vertical profiles of horizontal and time-averaged **a–c** total heat flux, **d–f** resolved heat flux, and **g–i** SGS heat flux, obtained with the D6 (left), D12 (middle), and D45 (right) grids from the precursor simulation for the SBL case using the different SGS models. Results from Gadde et al. (2021a) are added for comparison, together with a reference profile, denoted by $(D45)_{SGS}$, obtained by ensemble-averaging the profiles from all six SGS models with the D45 grid

Table 11 Values of the surface heat flux $\langle \bar{q}_w \rangle$ (K·m/s) and normalized Obukhov length L_O/z_h , obtained with the grids D6, D12, and D45, using the different SGS models

Model	$\langle \bar{q}_w \rangle$ (K·m/s)			L_O/z_h		
	D6	D12	D45	D6	D12	D45
Smag.	−0.0115	−0.0103	−0.0104	0.89	1.10	1.16
WALE	−0.0154	−0.0134	−0.0102	1.34	1.33	1.15
AMD	−0.0117	−0.0104	−0.0102	1.09	1.10	1.13
TKE	−0.0126	−0.0114	−0.0108	1.18	1.12	1.15
SDS	−0.0110	−0.0108	−0.0106	1.05	1.08	1.13
LASDD	−0.0212	−0.0157	−0.0113	1.56	1.37	1.27

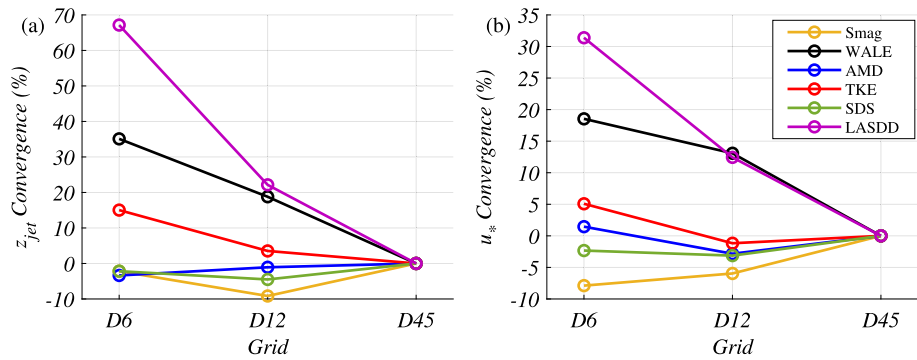


Fig. 19 Relative variation of **a** jet height z_{jet} and **b** friction velocity u_* obtained for D6 and D12 grids relative to values obtained using the D45 grid (i.e., $[(\cdot) - (\cdot)_{D45}] / (\cdot)_{D45}$) for the different SGS models

D Kinetic Energy Budget Analysis

The kinetic energy budget analysis is conducted, following the methodologies outlined by Allaerts and Meyers (2017) and Gadde and Stevens (2021b). This analysis involves examining the steady-state, time-averaged kinetic energy equation, expressed as follows:

$$\begin{aligned}
 \underbrace{\int_{\mathcal{V}} \rho_o f_i^{turb} u_i d\mathcal{V}}_{\text{Turbine power } (P)} &= \underbrace{\int_{\mathcal{V}} \rho_o \bar{u}_j \partial_j \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u_i' u_i'} \right) d\mathcal{V}}_{\text{Kinetic energy flux } (E_k)} \\
 &+ \underbrace{\int_{\mathcal{V}} \rho_o \partial_j \left(\frac{1}{2} \overline{u_j' u_i' u_i'} + \bar{u}_i \overline{u_j' u_i'} \right) d\mathcal{V}}_{\text{Turbulent transport } (T_t)} + \underbrace{\int_{\mathcal{V}} \rho_o \partial_j \left(\overline{u_i \tau_{ij}^{SGS}} \right) d\mathcal{V}}_{\text{SGS transport } (T_{SGS})} + \underbrace{\int_{\mathcal{V}} \rho_o \partial_i \left(\overline{P^* u_i} \right) d\mathcal{V}}_{\text{Flow work } (F)} \\
 &- \underbrace{\int_{\mathcal{V}} \rho_o g \beta \left(\overline{u_i \theta} - \bar{u}_i \bar{\theta}_o \right) \delta_{i3} d\mathcal{V}}_{\text{Buoyancy } (B)} + \underbrace{\int_{\mathcal{V}} \rho_o f \left(\bar{u}_1 U_{G2} - \bar{u}_2 U_{G1} \right) d\mathcal{V}}_{\text{Geostrophic forcing } (G)} - \underbrace{\int_{\mathcal{V}} \rho_o \overline{\tau_{ij}^{SGS} S_{ij}} d\mathcal{V}}_{\text{Dissipation } (D)},
 \end{aligned} \tag{54}$$

where $\overline{u_i' u_j'} = \overline{u_i u_j} - \bar{u}_i \bar{u}_j$. Each term in Eq. (54) is evaluated over a control volume $\mathcal{V}$ enclosing each wind-farm row (Fig. 20), where S_x and S_y are the normalized streamwise and spanwise turbine spacings, respectively. Importantly, the row-averaged power production is represented by the term $P_{row=n} = \int_{\mathcal{V}} \rho_o f_i^{turb} u_i d\mathcal{V}$, which denotes the average power of the n th row.

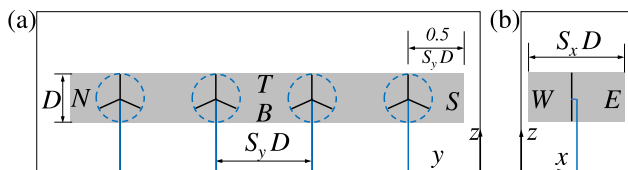


Fig. 20 Schematic showing **a** frontal and **b** lateral view of the control volume over which the integrals of the steady-state, time-averaged kinetic energy equation (Eq. (54)) are evaluated for one row of turbines. T, B, N, S, W, and E denote the top, bottom, north, south, west, and east surfaces of the control volume, respectively

Using the divergence theorem, the kinetic energy flux E_k can be expressed as the sum of its fluxes across the surfaces enclosing the control volume as:

$$\begin{aligned}
 & \underbrace{\int_V \rho_o \bar{u}_j \partial_j \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dV}_{\text{Kinetic energy flux } (E_k)} \\
 &= - \underbrace{\int_{S_W} \rho_o \bar{u} \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dydz}_{(E_{k,W})} + \underbrace{\int_{S_E} \rho_o \bar{u} \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dydz}_{(E_{k,E})} \\
 &\quad - \underbrace{\int_{S_S} \rho_o \bar{v} \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dx dz}_{(E_{k,S})} + \underbrace{\int_{S_N} \rho_o \bar{v} \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dx dz}_{(E_{k,N})} \\
 &\quad - \underbrace{\int_{S_B} \rho_o \bar{w} \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dx dy}_{(E_{k,B})} + \underbrace{\int_{S_T} \rho_o \bar{w} \left(\frac{1}{2} \bar{u}_i \bar{u}_i + \frac{1}{2} \overline{u'_i u'_i} \right) dx dy}_{(E_{k,T})}. \quad (55)
 \end{aligned}$$

Similarly, the turbulent transport T_i terms can be expressed as:

$$\begin{aligned}
 & \underbrace{\int_V \rho_o \partial_j \left(\frac{1}{2} \overline{u'_j u'_i u'_i} + \bar{u}_i \overline{u'_i u'_j} \right) dV}_{\text{Turbulent transport } (T_i)} \\
 &= - \underbrace{\int_{S_W} \rho_o \left(\frac{1}{2} \overline{u'_i u'_i u'_i} + \bar{u}_i \overline{u'_i u'_i} \right) dydz}_{(T_{i,W})} + \underbrace{\int_{S_E} \rho_o \left(\frac{1}{2} \overline{u'_i u'_i u'_i} + \bar{u}_i \overline{u'_i u'_i} \right) dydz}_{(T_{i,E})} \\
 &\quad - \underbrace{\int_{S_S} \rho_o \left(\frac{1}{2} \overline{v' u'_i u'_i} + \bar{u}_i \overline{u'_i v'} \right) dx dz}_{(T_{i,S})} + \underbrace{\int_{S_N} \rho_o \left(\frac{1}{2} \overline{v' u'_i u'_i} + \bar{u}_i \overline{u'_i v'} \right) dx dz}_{(T_{i,N})} \\
 &\quad - \underbrace{\int_{S_B} \rho_o \left(\frac{1}{2} \overline{w' u'_i u'_i} + \bar{u}_i \overline{u'_i w'} \right) dx dy}_{(T_{i,B})} + \underbrace{\int_{S_T} \rho_o \left(\frac{1}{2} \overline{w' u'_i u'_i} + \bar{u}_i \overline{u'_i w'} \right) dx dy}_{(T_{i,T})}, \quad (56)
 \end{aligned}$$

where the subscripts W , E , S , N , B , and T denote the flux across the west, east, south, north, bottom, and top boundaries, respectively (Fig. 20).

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Data Availability Data generated during the simulations can be requested from the authors or accessed on the figshare online repository: <https://doi.org/10.48420/28550888>.

Declarations

Conflict of interest The authors declare no conflict of interest.

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